

WHAT MATTERS FOR REPRESENTATION ALIGNMENT: GLOBAL INFORMATION OR SPATIAL STRUCTURE?

Anonymous authors
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ABSTRACT

Representation alignment (REPA) guides generative training by distilling representations from a strong, pretrained vision encoder to intermediate diffusion features. We resolve a fundamental question that has remained open in this literature: what aspect of the target representation actually matters for generation, its *global* semantic information (e.g., measured by ImageNet-1K accuracy) or its spatial structure (i.e. pairwise cosine similarity between patch tokens)? The prevailing wisdom holds that stronger global semantic performance makes for a better target representation. We show this belief is mistaken. Through a large-scale empirical analysis spanning 27 vision encoders and three model scales, we establish a clear conclusion: it is spatial structure, not global performance, that drives the generation performance of a target representation. We then turn this insight into a concrete method: two straightforward modifications that directly accentuate the transfer of *spatial* information. We replace the standard MLP projection layer in REPA with a simple convolution layer and introduce a spatial normalization layer for the external representation. Despite requiring fewer than four lines of code, our method, termed iREPA, delivers consistent improvements in the convergence speed of REPA across a diverse set of vision encoders, model sizes, and training variants (REPA, REPA-E, Meanflow, JiT, and more). Together, these results overturn the standard account of representation alignment and demonstrate how its true working mechanism can be leveraged for markedly more effective training of generative models.

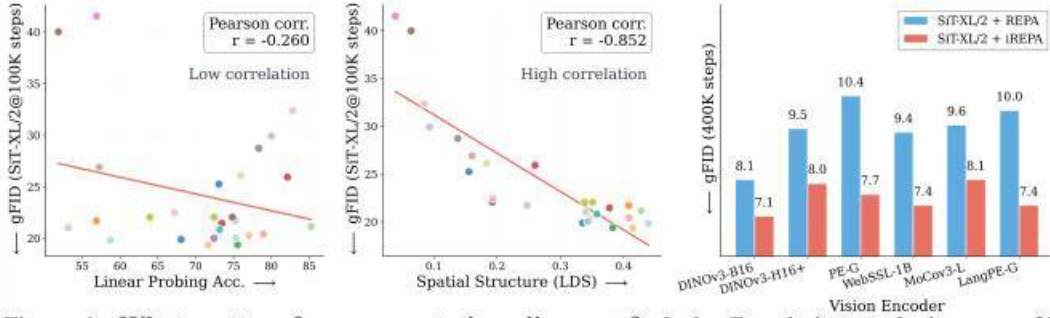


Figure 1: **What matters for representation alignment?** *Left:* Correlation analysis across 27 diverse vision encoders. Contrary to the prevailing wisdom, spatial structure, rather than global performance (measured by linear probing accuracy), is what drives the generation performance of a target representation. *Right:* Acting on this, we introduce two simple modifications that accentuate the transfer of spatial features from target representation to diffusion model. Our approach consistently improves the convergence speed of REPA across diverse settings.

1 INTRODUCTION

Representation alignment has become a powerful and widely adopted technique for accelerating the training of diffusion transformers (Ma et al., 2024; Peebles & Xie, 2023). By aligning internal diffusion representations with pretrained self-supervised visual encoders, recent methods deliver



Figure 2: **Motivating examples — spatial structure matters.** The comparison exposes a clear inverse relationship between ImageNet accuracy and generation quality. *Left:* PE-G, despite far higher validation accuracy (82.8% vs. 53.1%), generates *worse* than SpatialPE-B (Bolya et al., 2025). *Right:* WebSSL-1B (Fan et al., 2025) tells the same story: much better global performance (76.0% vs. 53.1%), but worse generation. **Spatial Self-Similarity:** Spatial structure, not global performance, is the reliable predictor of generation quality. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

substantial improvements in both convergence speed and final performance (Yu et al., 2024; Leng et al., 2025a). Yet for all these empirical successes, the field still lacks a clear account of *why* self-supervised features help diffusion model training. One fundamental question sits at the center of this gap: is the improvement driven by better global semantic information, as commonly measured through linear probing performance, or by better capturing spatial structure, characterized by the relationships between patch token representations? This paper answers that question decisively.

Getting this mechanism right is essential for advancing generative model training: it directly determines how one selects the optimal target representation and extracts its full benefit. The prevailing assumption today is that encoder performance for representation alignment tracks ImageNet-1K validation accuracy, a proxy for global semantic understanding (Oquab et al., 2024; Chen et al., 2021). Under this view, target representations with better ImageNet performance are expected to yield better generation (Yu et al., 2024). Accordingly, increases in linear probing accuracy of diffusion features are routinely cited as evidence for the effectiveness of representation alignment, cementing global semantic information as the presumed primary driver of improvement. The following quote from REPA (Yu et al., 2024) states this prevailing view directly:

“When a diffusion transformer is aligned with a pretrained encoder that offers more semantically meaningful representations (i.e., better linear probing results), the model not only captures better semantics but also exhibits enhanced generation performance, as reflected by improved validation accuracy with linear probing and lower FID scores.”

In this paper, we overturn this conventional wisdom. Through a systematic, large-scale empirical analysis across diverse vision encoders, including recent large vision foundation models such as WebSSL (Fan et al., 2025), DINOv3 (Siméoni et al., 2025), perceptual encoders (Bolya et al., 2025), and C-RADIO (Heinrich et al., 2024), we establish three findings that reshape how the community should think about what drives representation alignment.

Higher validation accuracy does not imply a better representation for generation. Directly contradicting the prevailing assumption, vision encoders with higher global semantic performance, measured by ImageNet-1K linear probing accuracy, frequently *underperform* for generation. Consider PE-Spatial-B, a spatially-tuned model derived from PE-Core-G (Bolya et al., 2025). Despite a far lower validation accuracy¹ on patch tokens than PE-Core-G (53.1% vs. 82.8%), PE-Spatial-B yields clearly better generation with REPA (Figure 2). WebSSL-1B (Fan et al., 2025) tells the same story: much stronger global performance (76.0% vs. 53.1%), yet worse generation. In Section §2 we show that this pattern is not anecdotal but holds across a wide range of target representations, pointing to a fundamental principle governing how representation alignment benefits diffusion training.

Spatial structure, not global information, determines generation performance. To pin down what actually matters, we introduce several straightforward metrics that measure the spatial self-similarity structure (Shechtman & Irani, 2007) between patch tokens (§3). We then run a large-scale analysis correlating generation FID under REPA against both linear probing accuracy and spatial structure,

¹Similar to REPA (Yu et al., 2024), we use linear probing accuracy on patch tokens to measure global semantic performance of external representation as only patch tokens are used for representation alignment.

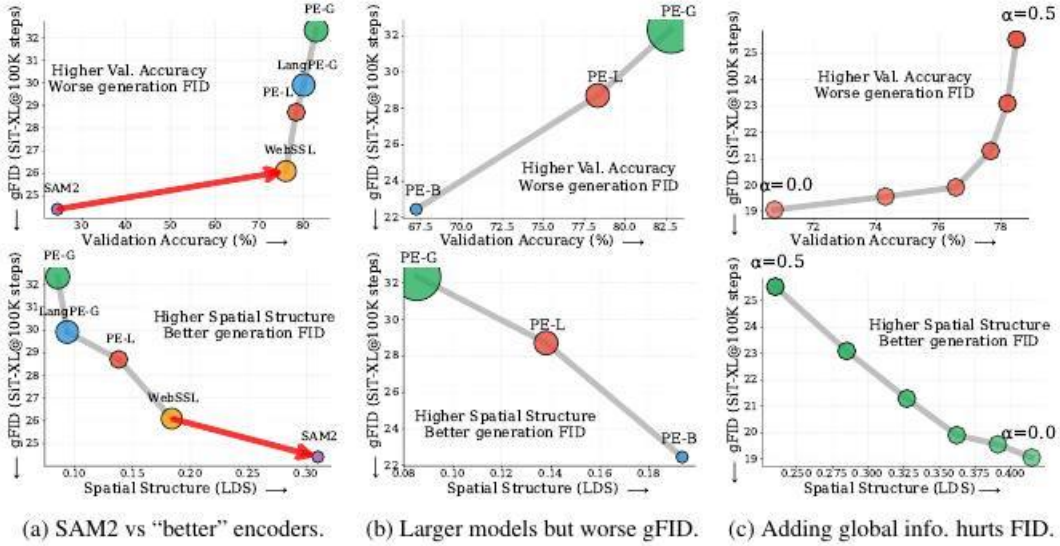


Figure 3: **Higher global information does not imply better REPA performance. Top row:** Three independent trends, all showing that global performance fails to track generation FID under REPA. (a) SAM2-S, with a validation accuracy of only 24.7%, generates better with REPA than models carrying $\sim 60\%$ higher validation accuracy. (b) Larger encoders within a family gain validation accuracy yet lose generation performance. (c) Adding global information to patch tokens via the CLS token raises global performance but hurts generation. **Bottom row:** Spatial structure, not global performance, is the reliable indicator of generation quality. See §3 for the detailed large-scale analysis across spatial structure metrics. All results are reported w/o classifier free guidance, SiT-XL/2 w/ REPA and 250 NFE (Yu et al., 2024) for inference.

across 27 vision encoders and 3 model sizes (SiT-B, SiT-L, SiT-XL). The result is unambiguous: every spatial structure metric correlates strongly with generation FID (Pearson $|r| > 0.852$), dwarfing the predictive power of ImageNet-1K validation accuracy ($|r| = 0.26$). Controlled experiments drive the point home: external representations carrying very limited global information already deliver significant gains with REPA. SAM2, for example, yields better generation with REPA than encoders with $\sim 60\%$ higher ImageNet accuracy (Fig. 3). Even classical spatial features such as SIFT (Lowe, 1999) and HOG (Dalal & Triggs, 2005), though not the strongest, achieve meaningful gains with representation alignment (§3).

Accentuating spatial features improves convergence. We turn this understanding into action with two straightforward modifications (<4 lines of code) that directly accentuate the transfer of “spatial” information from target representation to diffusion model: (1) a spatial regularization layer that boosts the spatial contrast of the target representations; and (2) a fix for the standard MLP-based projection layer (used to map diffusion features to target representation dimensions), which we show discards spatial information (Figure 6a) and which we replace with a simple convolution layer. The payoff is clear and consistent: our method, termed iREPA, reliably improves the convergence speed of REPA across diverse encoders, model sizes, and training recipes, including REPA-E (Leng et al., 2025a) and Meanflow (Geng et al., 2025) w/ REPA.

We summarize the main contributions of this paper below:

- We establish, through large-scale empirical analysis, that spatial structure and not global semantic information is what drives the effectiveness of representation alignment.
- We introduce a straightforward and fast-to-compute Spatial Structure Metric (SSM) that correlates far more strongly with downstream FID performance than linear probing scores, providing a reliable predictor where the standard proxy fails.
- We propose simple modifications that accentuate the transfer of spatial information from the target representation to diffusion features. Our method delivers consistent improvements in convergence speed over REPA across variations in target representation, model architecture, and training recipe (REPA, REPA-E, Meanflow w/ REPA, JiT w/ REPA, and more).

2 MOTIVATION: GLOBAL INFORMATION MATTERS LESS

We begin with a series of concrete examples that dismantle a core assumption: that a target representation with higher global performance (ImageNet-1K accuracy) implies better generation with REPA. It does not. In §3 we go further and show that these same observations, previously unexplained, are cleanly accounted for by the spatial structure of the target representation.

Recent vision encoders. We examine a range of recent vision encoders, including Perceptual Encoders (Bolya et al., 2025), WebSSL (Fan et al., 2025), and DINOv3 (Siméoni et al., 2025). Consider PE-Spatial-B (80M), a small spatially tuned model derived from PE-Core-G (1.88B) (Bolya et al., 2025). As Figure 2 shows, PE-Core-G attains far higher ImageNet-1K accuracy (82.8% vs 53.1%), yet performs distinctly worse as the target representation for REPA (FID 32.3 vs 21.0). WebSSL-1B (1.2B) repeats the pattern: much higher ImageNet-1K accuracy (76.0% vs 53.1%), yet worse generation under REPA (FID 26.1 vs 21.0).

SAM2 outperforms vision encoders with much higher ImageNet-1K accuracy. To test just how little global information generation actually requires, we analyze the SAM2-S vision encoder (46M) (Ravi et al., 2024) — a small model carrying very little global information, with a validation accuracy of only 24.1% (Fig. 3a). The result is striking: used for REPA, SAM2-S achieves better FID than vision encoders with dramatically higher ImageNet-1K accuracy, including PE-Core-G (82.8%).

Larger models within the same encoder family can have similar or worse generation performance. It is widely assumed that larger models within an encoder family carry better representations (as measured by ImageNet-1K accuracy). For representation alignment this assumption breaks down: larger variants often yield similar (DINOv2) or outright *worse* (PE, Cradio) generation performance (Fig. 3b). (Yu et al., 2024) observe the same effect for DINOv2 but can only speculate about its cause, attributing it to “all DINOv2 models being distilled from the DINOv2-g model and thus sharing similar representations”. We later provide the actual explanation, grounded in spatial structure (§3).

Adding more global information hurts generation. To directly test whether additional global information helps representation alignment, we run controlled experiments that inject global semantics, via the CLS token, into local patch tokens (with DINOv2 as encoder). The CLS token mixing operation is $\mathbf{p}_i^{\text{new}} = \mathbf{p}_i + \alpha \cdot \mathbf{c}$, where $\mathbf{p}_i$ denotes the i^{th} patch token, $\mathbf{c}$ the CLS token, and $\alpha \in [0, 0.5]$ controls the mixing strength. The two metrics move in opposite directions (Figure 3c): as α increases from 0 to 0.5, linear probing accuracy climbs monotonically from 70.7% to 78.5%, while generation quality degrades sharply, with FID worsening from 19.2 at $\alpha = 0$ to 25.4 at $\alpha = 0.5$. Adding global information does not merely fail to help; it actively harms generation.

Taken together, these observations establish that the global performance of a representation is simply not a reliable indicator of its REPA performance. We now show (§3) that spatial structure provides exactly the signal that global performance lacks.

- **Finding 1.** *Better global semantic information (e.g., ImageNet-1K accuracy) is not a reliable indicator of a better representation for generation, and often accompanies worse generation.*

3 SPATIAL STRUCTURE MATTERS MORE

Our central claim is that spatial structure, rather than global information, drives the generation performance of a target representation. To make this claim testable, we introduce several straightforward metrics that quantify the spatial self-similarity structure (Shechtman & Irani, 2007) of target representations. We then demonstrate two things: every spatial structure metric correlates far more strongly with gFID than ImageNet-1K accuracy does, and these metrics precisely explain the observations left unexplained in §2.

Measuring spatial self-similarity structure. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we define:

- **Self-similarity.** Let $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ be the extract patch representations, with $T = H \times W$ patches. Let kernel $K_{\mathcal{X}}(\cdot, \cdot) \in \mathbb{R}$ measure appearance self-similarity (Shechtman & Irani, 2007) between patch tokens. Here, we use the cosine kernel $K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2}$.

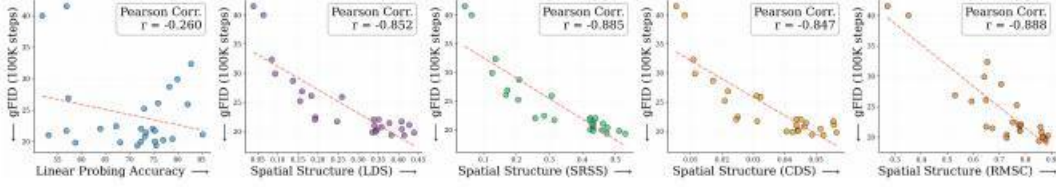


Figure 4: Spatial structure correlates far more strongly with generation quality than linear probing. Correlation analysis across 27 diverse vision encoders, SiT-XL/2 and REPA. Linear probing is only weakly predictive of FID (Pearson $|r| = 0.260$), whereas every spatial structure metric, LDS ($|r| = 0.852$), SRSS ($|r| = 0.885$), CDS ($|r| = 0.847$), and RMSC ($|r| = 0.888$), is a strong predictor of generation performance. See Fig. 10 for detailed plots with encoder labels.

- *Spatial distance.* Let $\mathcal{P} \in \mathbb{N}^{T \times 2}$ be the spatial location of each of the T tokens in coordinate space and $d(\cdot, \cdot) \in \mathbb{N}$ be the Manhattan distance between pairs of tokens.

We then measure *spatial* self-similarity metric as how self-similarity $K_{\mathcal{X}}$ varies with lattice distance d between patch tokens. Intuitively, larger values indicate stronger spatial organization (closer patches more similar to each other than patches further away). By default, we use a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}, \mathcal{P}) = \mathbb{E}_{d(t, t') \in (0, r_{\text{near}})} K_{\mathcal{X}}(t, t') - \mathbb{E}_{d(t, t') \geq r_{\text{far}}} K_{\mathcal{X}}(t, t').$$

The final spatial self-similarity score (LDS) is computed as the expectation over patch representation $\mathcal{X}$. We use $r_{\text{near}} = r_{\text{far}} = H/2$ here, though we found correlation to be robust to their exact choices. Please see Appendix B for exact details and alternative spatial metrics (CDS, SRSS, RMSC) which also perform effectively (see Fig. 4).

Spatial structure correlates far more strongly with generation performance than linear probing.

Our large-scale correlation analysis across 27 diverse vision encoders makes the gap impossible to ignore (Figure 4): the standard linear probing proxy correlates only weakly with generation (Pearson $|r| = 0.26$), whereas every SSM metric correlates strongly (Pearson $|r| > 0.85$).

The result holds across model scales. This is not an artifact of a single model size. Across SiT-B, SiT-L, and SiT-XL (Figure 5), linear probing remains weakly correlated with generation ($|r| < 0.306$), while spatial structure remains strongly correlated throughout ($|r| > 0.826$).

- **Finding 2.** *Spatial structure correlates far more strongly with generation performance than linear probing. We next use spatial metrics to explain trends that global performance leaves unexplained.*

Spatial structure explains the trends global performance cannot. As established in §2, global performance (ImageNet-1K accuracy) is not a predictive measure of effectiveness for representation alignment. Spatial structure is, and it accounts for each of the anomalies below.

(1) *PE-Spatial-B vs PE-Core-G:* Figure 2 shows PE-Core-G with far higher ImageNet-1K accuracy (82.8% vs 53.1%) but worse generation under REPA (FID 32.3 vs 22.0). Spatial structure resolves the apparent paradox at once: despite its lower global performance, PE-Spatial-B has markedly better spatial structure than PE-Core-G, which is precisely why it generates better.

(2) *SAM2 outperforms “better” vision encoders:* §2 shows SAM2, with a much lower ImageNet-1K accuracy (24.1%), outperforming encoders with $\sim 60\%$ higher accuracy. Fig. 3a shows these gains follow directly from SAM2’s stronger spatial structure.

(3) *Larger models in the same encoder family underperform:* As Fig. 3b shows, larger models in an encoder family gain ImageNet-1K accuracy but can lose spatial structure, and it is this loss that drives their worse generation under REPA. This explains the effect Yu et al. (2024) could only observe, namely that larger models can yield similar or worse generation.

(4) *Adding global information to patch tokens via the CLS token hurts generation:* In Figure 3c, mixing the CLS token into patch tokens raises global performance but suppresses spatial contrast, causing patch tokens to become highly similar to otherwise unrelated tokens (e.g., foreground object to background). The resulting loss of spatial structure is exactly what degrades generation.

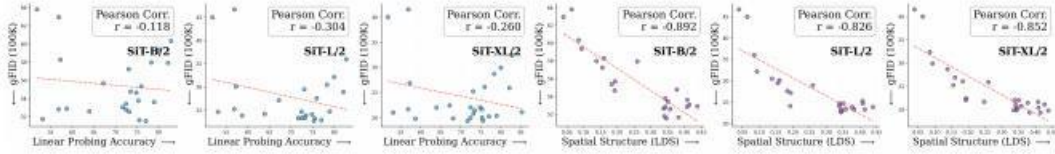
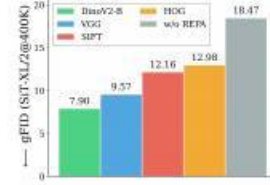
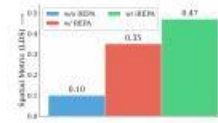


Figure 5: **Correlation analysis across model scales.** At every model scale, spatial structure (right) is far more predictive of gFID than linear probing (left), confirming that the effect is not specific to any single model size.

If spatial structure matters more, can we use SIFT or HOG features for REPA? Yes. Classical spatial features like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005) and intermediate VGG features (Simonyan & Zisserman, 2014), though not the strongest targets, all produce genuine performance gains with REPA. This is direct evidence that representation alignment benefits from spatial features alone, with no need for additional global information.



Can we use spatial metrics to explain gains with REPA? Yes. The spatial metrics we introduce explain both the gains from REPA itself and those from our improved training recipe (iREPA), which we present next in §4.



4 iREPA: IMPROVING REPRESENTATION ALIGNMENT BY ACCENTUATING WHAT MATTERS

Having established that spatial structure is what matters, we now act on it. We introduce two straightforward modifications to the original REPA training recipe, each designed to strengthen the transfer of spatial features from the teacher (vision encoder) to the student (diffusion transformer) model.

Convolutional projection layer instead of MLP. The standard REPA approach uses a 3-layer MLP projection to map diffusion feature dimensions to those of the external representation. This choice is actively harmful: as Figure 6a shows, the MLP projection is lossy and erodes the spatial contrast between patch tokens. We therefore replace it with a lightweight convolutional layer (kernel size 3, padding 1) operating directly on the spatial grid. The convolutional inductive bias preserves local spatial relationships by construction.

Spatial normalization layer. In line with Siméoni et al. (2025), we find that the patch tokens of pretrained vision encoders carry a substantial global component, evidenced by the high linear probing scores for the mean of patch tokens (Figure 14). Figure 6b makes the trade-off explicit: mixing this global information (mean of patch tokens) into the patch tokens raises global performance but suppresses the spatial contrast between individual patch tokens, causing tokens (e.g., a foreground object) to become highly similar to otherwise unrelated tokens (e.g., background).

Our analysis in §3 points to a clear remedy: since spatial contrast is what matters, we deliberately discard this global component (the mean of patch tokens) to sharpen the contrast between patch tokens. This yields a stronger spatial signal (pairwise similarity between patch tokens) and, with it, better REPA performance. Concretely, we add a simple spatial normalization layer (Ulyanov et al., 2016) to the patch tokens of the target representation:

$$\mathbf{y} = \frac{\mathbf{x} - \gamma \mathbb{E}[\mathbf{x}]}{\sqrt{\text{Var}[\mathbf{x}] + \epsilon}},$$

where $\mathbf{x} \in \mathbb{R}^{B \times T \times D}$ represents the patch tokens, the expectation and variance are computed across the spatial dimension, and $\epsilon = 10^{-6}$ for numerical stability.

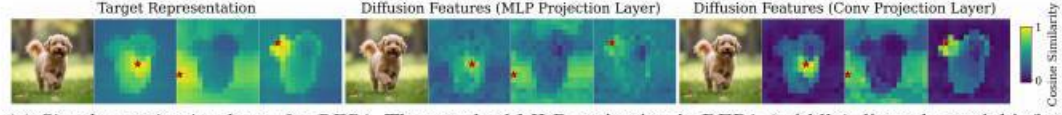
4.1 EXPERIMENTS

In this section, we establish the effectiveness of the improved training recipe through extensive experiments on Imagenet 256×256 . We organize the evaluation around the following research questions:

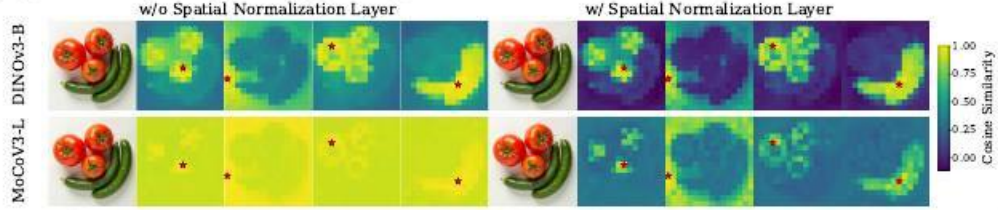
Algorithm 1 Summary of key iREPA modifications to standard REPA training

```
# 1. Conv projection instead of MLP
proj_layer = nn.Conv2d(D_in, D_out, kernel_size=3, padding=1)
```

```
# 2. Spatial normalization on encoder features [B, T, D]
x = x - gamma * x.mean(dim=1, keepdim=True)
x = x / (x.std(dim=1, keepdim=True) + 1e-6)
```



(a) *Simpler projection layer for REPA.* The standard MLP projection in REPA (middle) discards spatial information when transferring features from target representation (left) to diffusion features. A simpler convolution layer preserves it, yielding markedly better spatial information transfer (right).



(b) *Spatial normalization layer.* Patch tokens of pretrained vision encoders carry a global component that suppresses spatial contrast, causing tokens in one semantic region (e.g., tomato) to remain highly similar to unrelated tokens (e.g., background or cucumber). We deliberately discard this global information to sharpen the spatial contrast between patch tokens, which in turn improves generation performance.



(c) *Overall impact of improved training recipe (iREPA) on spatial structure of diffusion features.*

Figure 6: Two straightforward modifications (iREPA) that strengthen spatial feature transfer from target representation to diffusion features. All results reported with SiTL/2 with REPA at 100K.

- Can iREPA consistently improve the convergence speed of diffusion transformers over REPA across diverse external representations? (Figure 7, 8, 12, 13, 15)
- Is iREPA scalable in terms of model size and generalize across variations in training settings? (Table 1 [a,b,c], 2, 4, 5, 6, and, Figure 12, 16)
- Does iREPA generalize across more recent representation alignment methods such as REPA-E (Leng et al., 2025a), MeanFlow w/ REPA (Geng et al., 2025)? and pixel-space diffusion models such as JiT w/ REPA (Li & He, 2025b)? (Table 3 [a,b])

Convergence Speed. We evaluate the convergence behavior of iREPA across diverse vision encoders (DINOv3-B, WebSSL-1B, PE-Core-G, CLIP-L, MoCov3, PE-Lang-G) and model sizes (SiT-XL/2 and SiT-B/2). The results in Fig. 7 are unambiguous: iREPA achieves faster convergence than baseline REPA in every setting, across both target representation and model size.

Target representation. We analyze how iREPA generalizes across vision encoders in Fig. 8 and Table 4. iREPA improves generation quality for every vision encoder, without exception. Comprehensive comparisons across all 27 encoders are provided in Appendix C, E.

Encoder size. Table 1a analyzes how iREPA generalizes across encoder sizes: PE-B (90M), PE-L (320M), PE-G (1.88B). iREPA improves performance at every encoder size, and the gains grow with encoder size (22.2% for PE-B, 38.8% for PE-L, 39.6% for PE-G). Notably, this reverses the usual liability of larger encoders under REPA, turning them into the settings where our method helps most.

Scalability. Table 1b examines iREPA across model scales. The spatial improvements not only help at every scale, but the percentage gains *increase* with model size, demonstrating that our approach scales favorably rather than washing out on larger models.

Enc. Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
PE-B (90M)	59.6	22.5	6.23	0.63	0.60
+iREPA	73.3	17.5	6.04	0.66	0.62
PE-L (320M)	48.6	28.7	6.53	0.59	0.60
+iREPA	76.3	17.6	6.36	0.65	0.61
PE-G (1.88B)	42.7	32.3	6.70	0.57	0.59
+iREPA	70.8	19.5	6.35	0.64	0.61

(a) Vision Encoder size

Model Size	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
SiT-B	27.5	49.50	7.00	0.46	0.59
+iREPA	34.1	43.37	7.87	0.50	0.60
SiT-L	55.7	24.10	6.25	0.62	0.60
+iREPA	66.9	20.28	6.14	0.63	0.62
SiT-XL	70.3	19.06	5.83	0.65	0.61
+iREPA	77.9	16.96	6.26	0.66	0.61

(b) Model size

Aln. Depth	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Pr. $\uparrow$	Re. $\uparrow$
Layer 4	28.1	49.0	6.83	0.46	0.60
+iREPA	41.7	36.0	6.61	0.52	0.62
Layer 6	26.9	50.6	7.00	0.46	0.59
+iREPA	42.4	35.3	6.90	0.53	0.62
Layer 8	24.7	54.8	7.35	0.44	0.58
+iREPA	38.3	38.9	7.19	0.51	0.62

(c) Encoder depth (SiT-B/2)

Table 1: **Variation in training settings.** iREPA delivers consistent gains over baseline REPA across every training setting tested. Unless otherwise specified all results are reported using Dinov2 as encoder, SiT-XL/2, 100K steps and vanilla-REPA as baseline.

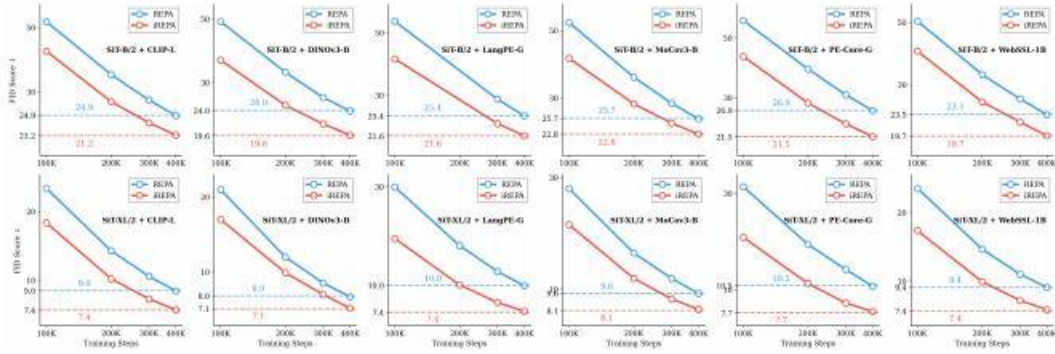


Figure 7: **Accelerating spatial features consistently improves convergence speed.**

Encoder depth. Table 1c analyzes iREPA across alignment depths (SiT-B/2 at 100K iterations, DINOv3-B as target representation). iREPA improves over baseline REPA at every alignment depth tested.

Abalation on different components. Table 2 isolates the contribution of each component of iREPA. Both the spatial normalization layer and the convolution projection layer individually deliver significant gains over baseline REPA, and combining them is best, confirming that the two modifications are complementary.

Training recipe. Finally, we test whether iREPA generalizes beyond vanilla REPA to recent training recipes such as REPA-E (Leng et al., 2025a) and MeanFlow w/ REPA (Geng et al., 2025). It does: Table 3 shows that the spatial improvements yield convergence gains under every recipe, indicating that they capture a mechanism common to representation alignment rather than a quirk of one recipe.

Classifier-free guidance. Table 4 reports generation quality with CFG across different vision encoders at 400K training iterations. iREPA delivers faster convergence for every encoder, both with and without classifier-free guidance, so the gains are not an artifact of the sampling procedure.

Pixel-space diffusion (JiT). We further push iREPA to pixel-space diffusion models such as JiT-B (Li & He, 2025b). Figure 9 compares REPA and iREPA under JiT (Li & He, 2025b): accentuating the transfer of spatial information again yields faster convergence across vision encoders (e.g., DiNOv2, DiNOv3, PE, etc.), confirming that our findings extend beyond latent-space diffusion.

- **Finding 3.** *Accentuating the transfer of spatial structure from target representation to diffusion features consistently boosts convergence speed with representation alignment (REPA), across encoders, model sizes, and training recipes.*

5 RELATED WORK

We discuss the most relevant related work here and provide a detailed discussion in Appendix H.

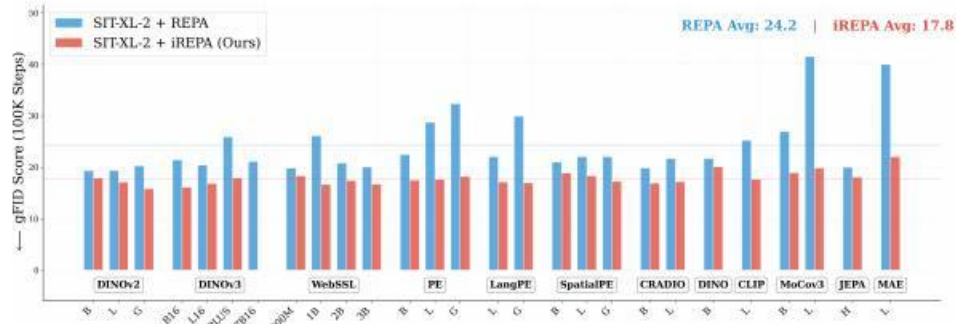


Figure 8: **Variation in target representation.** Across all 27 vision encoders, accentuating the transfer of spatial features from target representation to diffusion features (iREPA) consistently improves convergence speed. See Appendix C for more results across diverse settings.

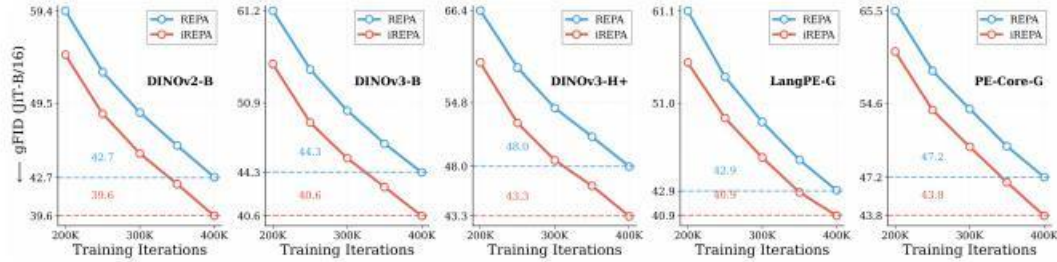


Figure 9: **Convergence comparison with pixel-space diffusion (JiT).** FID convergence curves comparing REPA vs iREPA with pixel-space JiT (Li & He, 2025b) across different vision encoders. Accentuating the transfer of spatial structure consistently improves convergence speed across vision encoders for pixel-space diffusion models as well. All results are reported with JiTB/16 (Li & He, 2025b), 256 batch size and without classifier-free guidance. Refer Table 10 for further results.

Representation alignment for generation. A growing body of work uses external representations to improve diffusion model training (Pernias et al., 2023; Fuest et al., 2024). In particular, recent methods (Yu et al., 2024; Yao & Wang, 2025; Leng et al., 2025a;b; Kouzelis et al., 2025) obtain significant gains by aligning internal diffusion features with clean image features from a pretrained vision encoder, and this idea has been extended to video generation (Zhang et al., 2025), 3D generation (Wu et al., 2025), and unified models (Ma et al., 2025a). What all of these successes share is an unexplained mechanism: it has remained unclear precisely how self-supervised features enhance diffusion training. We close this gap, identifying *which* aspect of the target representation matters for generation and turning that answer into an improved training recipe.

Spatial vs global information tradeoff in pretrained vision encoders. A related line of work studies the tradeoff between global and spatial information in pretrained vision encoders (Bolya et al., 2025; Siméoni et al., 2025). (Siméoni et al., 2025) show that continued self-supervised training can increase the similarity between the global CLS token and patch tokens, degrading performance on dense spatial tasks, while (Bolya et al., 2025; Heinrich et al., 2024) explicitly train spatial-tuned models for dense prediction. We establish that this same tradeoff is decisive for generation: the spatial structure of a vision encoder matters more than its global information. This reframes how external representations should be selected and trained for generative tasks, and we expect it to guide substantial future work in this direction.

6 CONCLUSION

This paper settles what truly drives the effectiveness of representation alignment: global information or the spatial structure of the target representation. Through large-scale empirical analysis we establish a clear and, at first, counterintuitive answer: it is spatial structure, not global information,

Method	DINOv2-B			DINOv3-B			WebSSL-1B			PE-Core-G		
	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓	FID↓	IS↑	sFID↓
Baseline REPA	19.06	70.3	5.83	21.47	63.4	6.19	26.10	53.0	6.90	32.35	42.7	6.70
iREPA (w/o spatial norm)	18.52	73.3	6.11	17.76	74.7	5.81	21.17	64.6	6.27	24.97	57.4	6.21
iREPA (w/o conv proj)	17.66	72.8	6.03	18.28	70.8	6.18	18.44	71.0	6.22	21.72	61.5	6.26
iREPA (full)	16.96	77.9	6.26	16.26	78.8	6.14	16.66	77.5	6.18	18.19	75.0	6.03

Table 2: **Ablation on different components.** Across every encoder, both the spatial normalization layer and the convolution projection layer (§4) deliver significant gains in convergence speed, and combining them is consistently best, confirming the two are complementary. All results reported at 100K steps and SiT-XL/2.

Encoder	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
WebSSL-1B	52.8	26.5	5.20	0.620	0.585
+iREPA-E	87.0	13.2	5.28	0.699	0.598
PE-G	50.9	25.9	5.68	0.612	0.576
+iREPA-E	80.0	16.4	5.40	0.667	0.616
DINOv3-B	82.2	14.4	4.68	0.694	0.596
+iREPA-E	93.6	11.7	4.57	0.703	0.613
DINOv2-B	87.5	12.9	5.40	0.708	0.586
+iREPA-E	91.3	12.1	4.86	0.712	0.602

(a) REPA-E (SiT-XL/2)

Encoder	w/o CFG				w/ CFG			
	4 NFE		1 NFE		4 NFE		1 NFE	
	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓	IS↑	FID↓
WebSSL-1B	27.22	51.40	24.14	58.67	87.85	16.59	69.06	23.74
+iREPA	31.48	45.67	27.33	55.70	100.74	13.89	78.67	20.69
DINOv3-B	28.36	49.58	25.47	57.01	93.30	15.56	72.41	22.55
+iREPA	33.63	44.52	29.67	53.75	124.54	11.05	98.93	17.32
DINOv2-B	28.77	48.87	25.43	56.63	94.40	15.28	71.95	22.17
+iREPA	33.63	44.52	29.67	53.75	111.44	12.59	90.25	18.62

(b) MeanFlow w/ REPA (SiT-B/2)

Table 3: **Variation in training recipes.** Applied on top of REPA-E (a) and Meanflow with REPA (b), our spatial improvements yield consistent gains under both recipes. For Meanflow, we report results at 1 and 4 NFEs. CFG value of 2.0 is used for Meanflow. All results are reported at 100K training iterations.

that drives representation alignment. Acting on this, we introduce two simple modifications that accentuate the transfer of spatial information from target representation to diffusion features. The resulting method, iREPA, consistently improves the convergence speed of REPA across diverse vision encoders and training recipes. More broadly, our results overturn the standard account of why representation alignment works and lay out a concrete, actionable basis for leveraging its true mechanism to train generative models more effectively.

Vision Encoder	Steps	w/o CFG					w/ CFG				
		IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	100K	69.20	19.3	5.89	0.64	0.61	157.2	6.35	5.91	0.769	0.536
+IREPA	100K	77.92	16.9	6.26	0.66	0.61	179.3	5.15	6.23	0.783	0.544
DINOv2-B	400K	127.4	7.76	5.06	0.70	0.66	263.0	1.98	4.60	0.799	0.610
+IREPA	400K	128.6	7.52	4.89	0.71	0.65	268.8	1.93	4.59	0.799	0.600
DINOv3-B	100K	63.64	21.4	6.14	0.63	0.60	144.0	7.57	6.09	0.762	0.526
+IREPA	100K	78.79	16.2	6.14	0.66	0.61	181.9	4.87	6.10	0.780	0.547
DINOv3-B	400K	126.7	8.10	5.06	0.70	0.66	261.2	1.99	4.58	0.799	0.609
+IREPA	400K	132.9	7.13	4.93	0.71	0.66	272.4	1.89	4.58	0.799	0.600
WebSSL-1B	100K	53.87	25.5	6.57	0.61	0.59	124.0	9.59	6.37	0.756	0.515
+IREPA	100K	77.47	16.6	6.18	0.66	0.61	177.6	5.09	6.11	0.787	0.538
WebSSL-1B	400K	116.9	9.39	5.14	0.70	0.64	250.5	2.24	4.61	0.809	0.580
+IREPA	400K	130.8	7.48	4.91	0.70	0.65	271.6	1.90	4.58	0.798	0.609
PE-Core-G	100K	42.74	32.3	6.70	0.57	0.59	97.2	14.1	6.56	0.714	0.525
+IREPA	100K	75.01	18.1	6.03	0.64	0.61	176.8	5.66	6.08	0.771	0.544
PE-Core-G	400K	109.4	10.4	5.00	0.69	0.64	238.4	2.44	4.57	0.805	0.585
+IREPA	400K	132.0	7.78	5.02	0.70	0.65	275.4	1.93	4.59	0.796	0.606

Table 4: **Results across different encoders with and w/o classifier-free guidance.** SiT-XL/2 is used base model. See Appendix E for detailed results on SiT-B/2 & SiT-L/2 across all encoders.

REPRODUCIBILITY STATEMENT

We provide all implementation details and hyperparameters in Appendix G. Our code, model checkpoints, and analysis results will be released upon acceptance.

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A DETAILED CORRELATION ANALYSIS

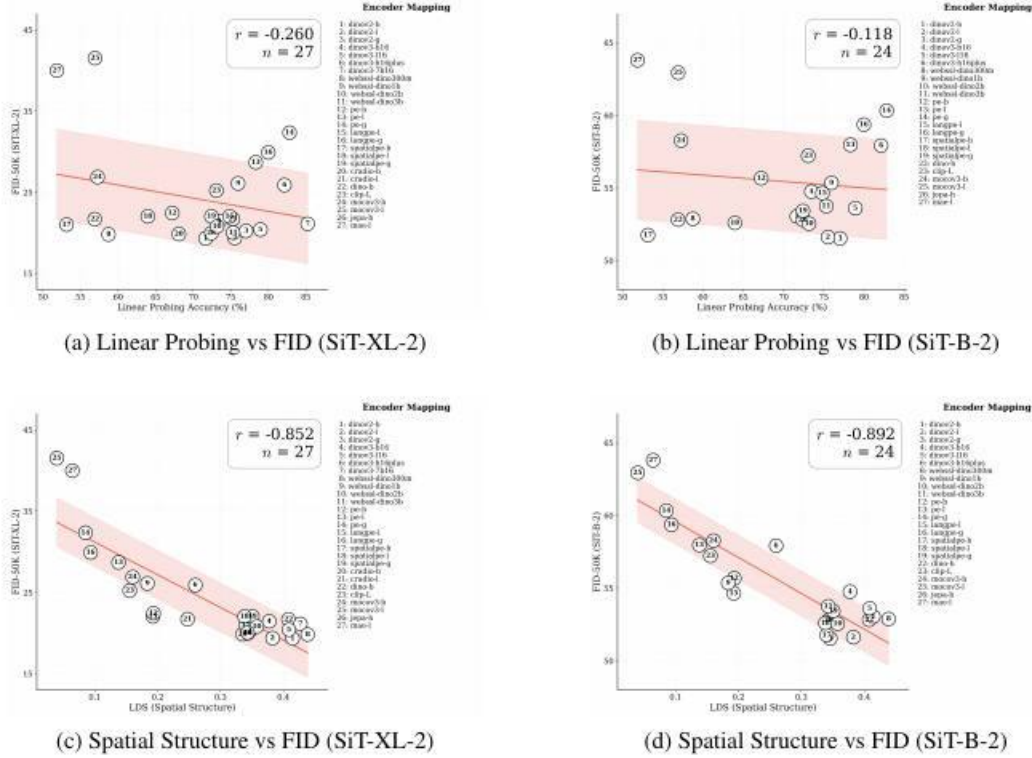


Figure 10: **Detailed correlation analysis with encoder labels.** Top row shows linear probing accuracy vs FID correlation for SiT-XL-2 and SiT-B-2 models. Bottom row shows spatial structure (LDS) vs FID correlation. Each point is labeled with its corresponding encoder number (see legend). The spatial structure metric is decisively the stronger predictor of generation quality at both model scales.

B SPATIAL SELF-SIMILARITY METRICS

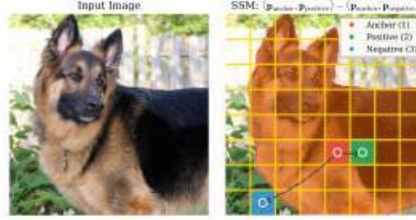


Figure 11: **Explaining semantic region self-similarity (SRSS) metric.** Visual explanation of SRSS metric: For each image, we compute a mask using SAM2 and select anchor-positive-negative triplets. The SSM measures the difference in cosine similarity between anchor-positive pairs (within mask) and anchor-negative pairs (outside mask). Intuitively, larger values indicate that patches on the same semantic region are more similar than patches on unrelated regions — indicating better spatial structure preservation.

Setup. Given an image $\mathcal{I}$ and vision encoder $\mathcal{E}$, we extract patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ with $T = H \times W$ and indices $t \in [T]$ placed on an $H \times W$ lattice. Let $d : [T] \times [T] \rightarrow \mathbb{N}$ be the Manhattan distance, and let the (cosine) self-similarity kernel be

$$K_{\mathcal{X}}(t, t') = \frac{\langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle}{\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2} \in [-1, 1],$$

a standard local self-similarity measure (Shechtman & Irani, 2007).

- *Patch token representation:* $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$ where $T = H \times W$ is the spatial grid of patches (patch index set $[T]$). Also let $d : [T] \times [T] \rightarrow \mathbb{N}$ be Manhattan distance on the $H \times W$ grid.
- *Self-similarity kernel:* $K : [T] \times [T] \rightarrow \mathbb{R}$ measuring self-similarity (Shechtman & Irani, 2007) between patch tokens. We use the cosine kernel $K_{\mathcal{X}}(t, t') = \langle \mathbf{x}_t, \mathbf{x}_{t'} \rangle / (\|\mathbf{x}_t\|_2 \|\mathbf{x}_{t'}\|_2)$.
- *Spatial self-similarity metric:* a functional $m : \mathcal{K} \times \mathcal{D} \rightarrow \mathbb{R}$ measures how self-similarity K between patch tokens varies with lattice distance d . Intuitively, larger values indicate stronger spatial organization (near patches more similar to each other than far away patches).

We now define several straightforward metrics for measuring the spatial self-similarity structure of the target representations; as we show, all of them are strong predictors of generation quality.

Local vs. Distant Similarity (LDS). We first consider a simple correlogram contrast (local vs. distant) metric (Huang et al., 1997):

$$\text{LDS}(\mathcal{X}) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') < r_{\text{near}}] - \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') \geq r_{\text{far}}].$$

where r_{near} and r_{far} are the hyperparameters. By default we use $r_{\text{near}} = r_{\text{far}} = H/2$. We found correlation to be robust to their exact choices. Intuitively, larger values indicate that *on average*, patches that are closer to each other are more similar than patches that are further away.

Correlation Decay Slope (CDS). Given the patch tokens $\mathcal{X} = \mathcal{E}(\mathcal{I}) \in \mathbb{R}^{T \times D}$, we compute the spatial correlogram $g_{\mathcal{X}}(\delta) = \mathbb{E}[K_{\mathcal{X}}(t, t') \mid d(t, t') = \delta]$ for distances $\delta \in \{1, \dots, \Delta\}$. We then fit a least-squares line $\hat{g}_{\mathcal{X}}(\delta) \approx \alpha + \beta\delta$ and define:

$$\text{CDS}(\mathcal{X}) = -\hat{\beta}$$

where $\hat{\beta}$ is the fitted slope. Larger CDS values indicate faster similarity decay with distance, suggesting stronger spatial organization.

Semantic-Region Self-Similarity (SRSS). For each image, we first sample a binary mask $\mathbf{M} \in \mathbb{R}^{H \times W}$ containing a random object using SAM2 (Ravi et al., 2024) and downsampled to $H \times W$. We then select three types of points: anchor, positive, and negative. The anchor and positive points are sampled from within the mask (should have similar semantics), while the negative point is sampled from outside the mask (less related to anchor). Conceptually, if the encoder feature representation preserves the spatial structure, the anchor and positive points should have higher cosine similarity,

while the anchor and negative points should have lower cosine similarity. Thus, we define the Spatial Structure Metric (SSM) as:

$$\text{SSM}(\mathbf{P}) = \mathbb{E}_{\text{anchor} \in \mathbf{M}} [\cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{pos}}) - \cos(\mathbf{p}_{\text{anchor}}, \mathbf{p}_{\text{neg}})]$$

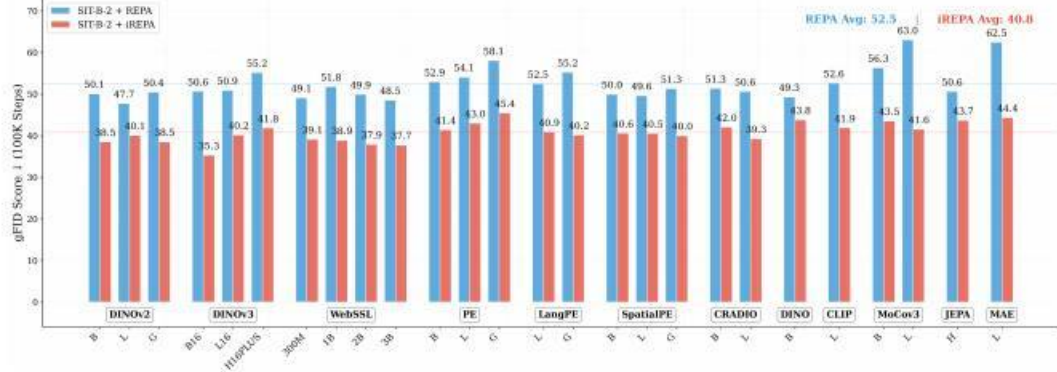
where positive points are within Manhattan distance $d \leq r_{\text{near}}$ from the anchor within the mask, and negative points are at distance $d \geq r_{\text{far}}$ outside the mask.

RMS Spatial Contrast (RMSC). Finally, we consider a simple contrast metric to capture the spatial contrast between patch token representations. Given normalized features $\hat{\mathbf{x}}_t = \mathbf{x}_t / \|\mathbf{x}_t\|_2$ for each patch t , we compute:

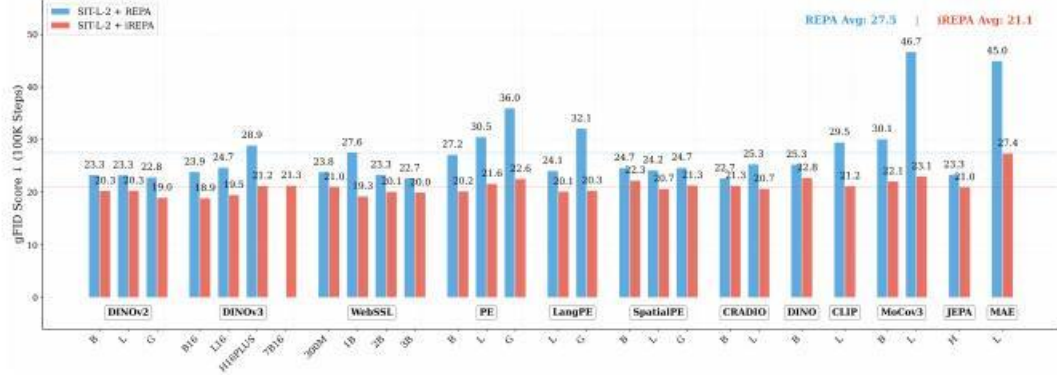
$$\text{RMSC}(\mathcal{X}) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{x}}_t - \bar{\mathbf{x}}\|_2^2}$$

where $\bar{\mathbf{x}} = \frac{1}{T} \sum_{t=1}^T \hat{\mathbf{x}}_t$ is the mean of normalized features across all spatial locations. Higher RMSC values indicate greater spatial diversity in the feature representations, suggesting preserved spatial structure, whereas lower values indicate more uniform feature distributions indicating a loss of spatial structure (typically happens global information diminishes spatial structure (Siméoni et al., 2025)).

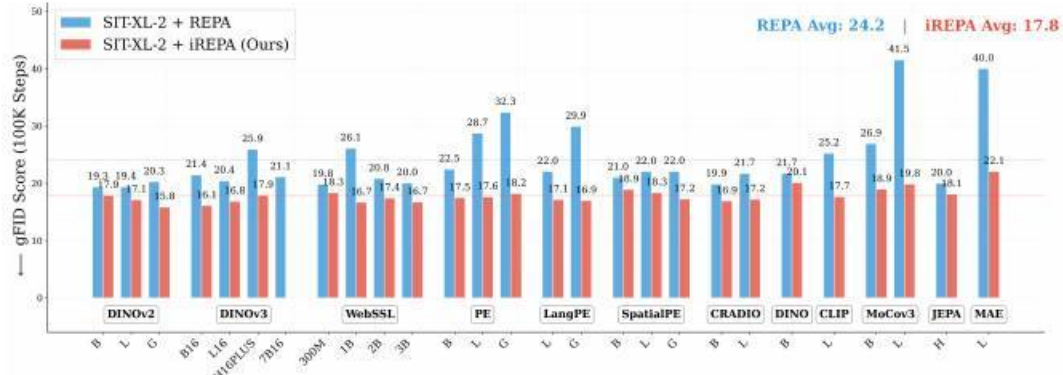
C VISION ENCODER VARIATION



(a) SiT-B-2 (130M parameters)



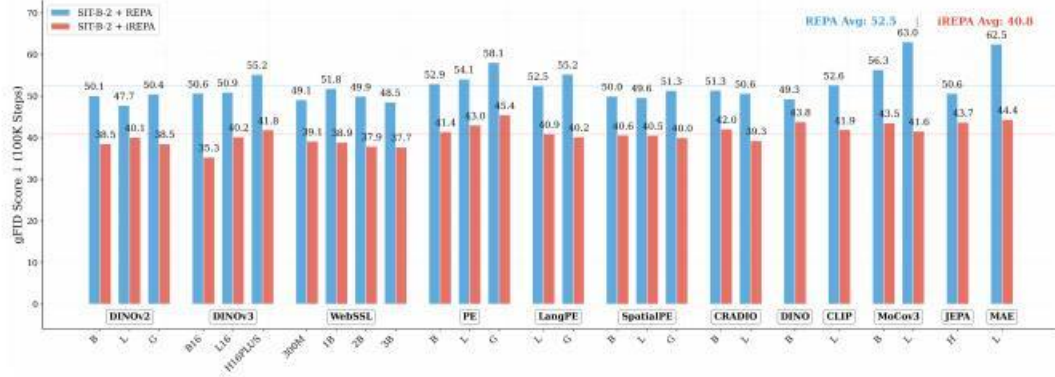
(b) SiT-L-2 (458M parameters)



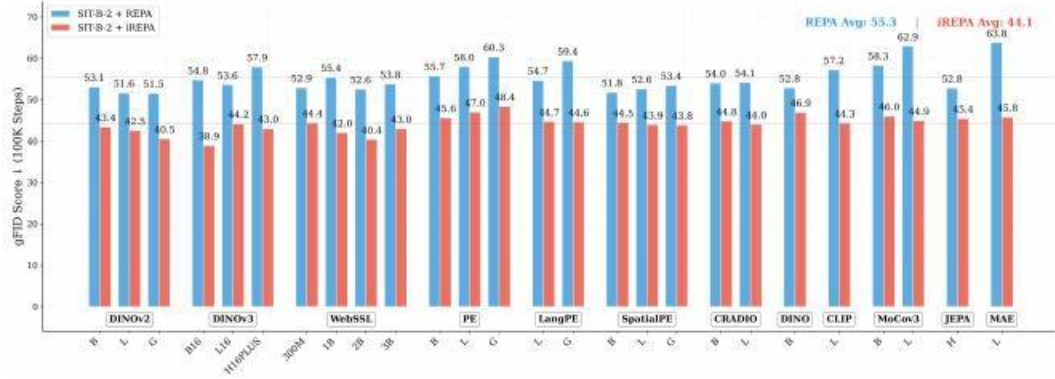
(c) SiT-XL-2 (675M parameters)

Figure 12: **Variation in vision encoders.** Ablation studies of iREPA across vision encoders for SiT-B-2, SiT-L-2, and SiT-XL-2. iREPA improves generation quality for every encoder and at every model size.

D ENCODER DEPTH VARIATION



(a) SiT-B-2 with 6 encoder layers



(b) SiT-B-2 with 8 encoder layers

Figure 13: Effect of encoder depth on iREPA performance. Comparison of SiT-B-2 performance with different encoder depths (6 vs 8 layers) across various vision encoders. iREPA improves performance regardless of encoder depth, with the 8-layer configuration performing slightly better. Both configurations show significant FID improvements once spatial regularization is applied.

E COMPREHENSIVE RESULTS ACROSS VISION ENCODERS

E.1 SiT-B/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	59.44	24.84	6.46	0.584	0.645
+iREPA	69.70	21.26	6.64	0.601	0.644
DINOv2-B	64.92	22.75	6.54	0.593	0.653
+iREPA	70.88	21.40	6.77	0.597	0.653
DINOv2-L	66.65	22.46	6.53	0.594	0.650
+iREPA	72.41	20.99	6.84	0.600	0.650
DINOv3-B16	62.57	23.91	6.77	0.584	0.647
+iREPA	75.48	19.65	6.68	0.606	0.649
DINOv3-H16+	61.91	24.82	6.57	0.579	0.645
+iREPA	71.76	21.80	6.84	0.592	0.655
DINO-B	58.37	24.71	6.32	0.586	0.635
+iREPA	57.55	24.41	6.43	0.587	0.638
LangPE-G	58.42	25.37	6.32	0.583	0.641
+iREPA	66.88	21.62	6.65	0.604	0.649
MoCov3-B	57.61	25.60	6.25	0.579	0.640
+iREPA	61.41	22.80	6.46	0.596	0.639
PE-B	58.86	25.78	6.44	0.58	0.64
+iREPA	69.50	21.15	6.66	0.601	0.659
PE-G	56.09	26.84	6.45	0.568	0.641
+iREPA	70.43	21.52	6.58	0.598	0.651
SpatialPE-B	59.19	24.28	6.37	0.586	0.646
+iREPA	61.50	22.71	6.41	0.604	0.644
SpatialPE-L	62.13	23.87	6.51	0.59	0.65
+iREPA	68.83	21.07	6.59	0.607	0.651
WebSSL-1B	63.20	23.48	6.33	0.593	0.646
+iREPA	73.54	19.74	6.63	0.612	0.646
WebSSL-2B	66.56	22.27	6.69	0.596	0.654
+iREPA	72.13	20.47	6.89	0.609	0.648
WebSSL-300M	62.06	23.64	6.48	0.590	0.646
+iREPA	66.83	21.46	6.71	0.606	0.653
WebSSL-3B	65.91	22.55	6.53	0.598	0.654
+iREPA	71.98	20.64	6.71	0.607	0.650

Table 5: **SiT-B/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality for every encoder. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

E.2 SiT-L/2 RESULTS

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
CLIP-ViT-L	107.7	10.6	5.20	0.682	0.643
+iREPA	115.8	9.41	5.15	0.689	0.653
DINOv2-B	113.3	9.61	5.11	0.684	0.653
+iREPA	113.9	9.36	5.11	0.690	0.655
DINOv3-B16	115.3	9.41	5.24	0.687	0.651
+iREPA	118.1	9.06	5.25	0.685	0.650
DINOv3-H16+	110.7	10.6	5.32	0.677	0.652
+iREPA	120.5	9.64	5.29	0.681	0.663
LangPE-G	102.8	11.2	5.13	0.684	0.637
+iREPA	112.7	9.37	5.02	0.690	0.653
MoCov3-B	102.1	11.0	5.14	0.685	0.635
+iREPA	104.7	10.1	5.00	0.690	0.641
PE-G	98.38	12.6	5.31	0.670	0.646
+iREPA	117.1	9.65	5.33	0.683	0.651
WebSSL-1B	107.9	10.5	5.13	0.686	0.644
+iREPA	121.5	8.62	5.01	0.696	0.653

Table 6: **SiT-L/2 performance across vision encoders at 400K iterations.** iREPA improves generation quality across all encoders, with especially strong gains in FID and IS. All baselines are reported using vanilla-REPA (Yu et al., 2024) for training.

F ADDITIONAL EXPERIMENTAL RESULTS

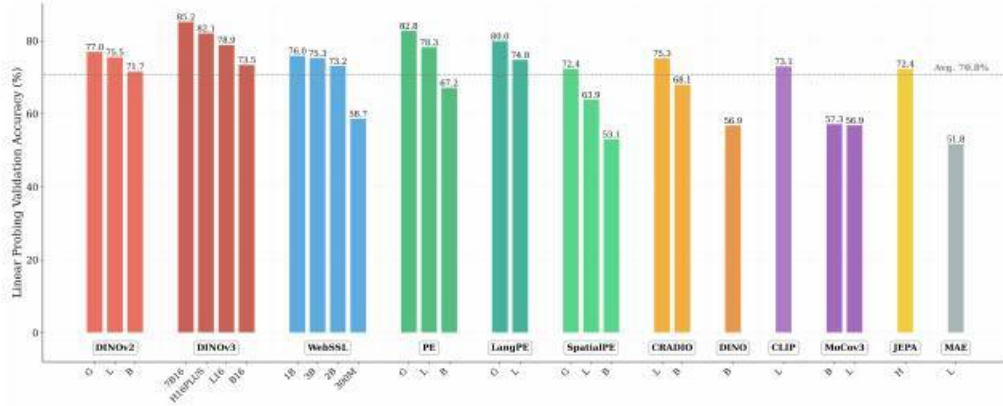


Figure 14: **Global information in mean patch tokens.** Beyond the [CLS] token, the mean of patch tokens itself carries substantial global semantic information. This raises global performance but suppresses the contrast between individual patch tokens, hindering spatial structure transfer. Removing this global information (via the mean of patch tokens) improves spatial structure transfer during representation alignment (§4).

Vision Encoder	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
DINOv2-B	262.97	1.98	4.60	0.799	0.610
+iREPA (Ours)	268.79	1.93	4.59	0.799	0.600
DINOv3-B	261.18	1.99	4.58	0.799	0.609
+iREPA (Ours)	272.41	1.89	4.58	0.799	0.600
WebSSL-1B	250.53	2.24	4.61	0.809	0.580
+iREPA (Ours)	271.59	1.90	4.58	0.798	0.609
PE-G	238.37	2.44	4.57	0.805	0.585
+iREPA (Ours)	275.36	1.93	4.59	0.796	0.606

Table 7: **Generation quality with classifier-free guidance.** Comparison of REPA vs iREPA with CFG (scale 2.0) across different vision encoders. All experiments use SiT-XL/2 trained for 400K iterations with 250 sampling steps. iREPA improves both IS and FID for every encoder, confirming the gains persist under CFG.

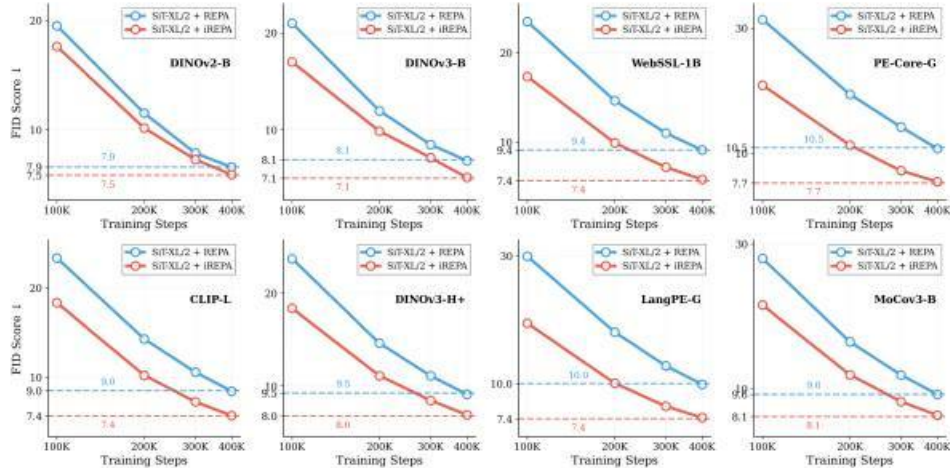


Figure 15: **Accentuating spatial features helps consistently improve convergence speed.** Results for SiT-XL/2 with REPA and iREPA.

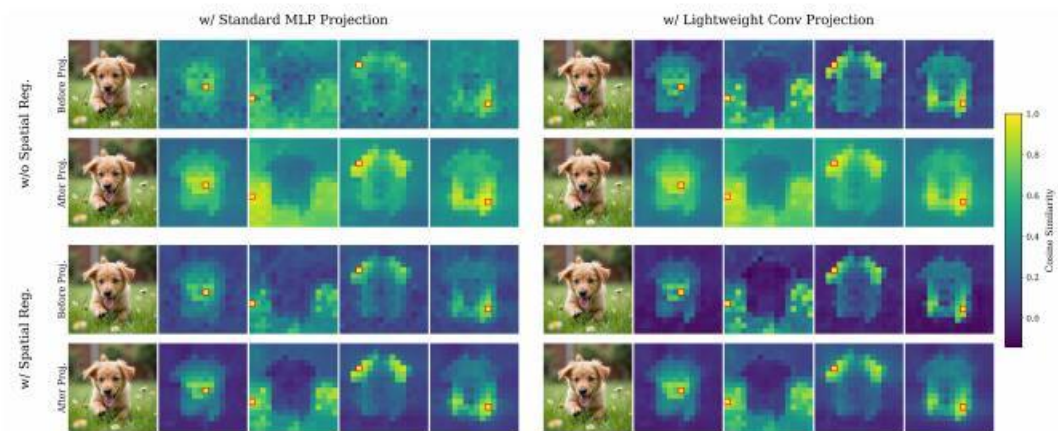


Figure 16: **Visualizing the impact of two straightforward improvements to enhance spatial feature transfer.** **First**, the standard MLP projection layer (top-left) discards spatial information when transferring features from the pretrained encoder (after projection) to the diffusion model (before projection). A simpler convolution layer preserves this spatial information (top-right). **Second**, vision encoder features often have limited spatial contrast (§2), causing tokens in one semantic region (*e.g.*, dog) to remain highly similar to unrelated tokens (*e.g.*, background). We fix this with a simple spatial regularization layer that sharpens the spatial contrast in the learned representation, directly improving generation performance. Combining both modifications is best (refer Table 2).

G IMPLEMENTATION DETAILS

Training setup. We follow the same setup as in REPA (Yu et al., 2024) unless otherwise specified. All training is conducted on the ImageNet (Deng et al., 2009) training split. For preprocessing, we adopt the protocol from ADM (Dhariwal & Nichol, 2021), where each image is center-cropped and resized to 256×256 resolution. We use `stabilityai/sd-vae-ft-mse` VAE (AI, n.d.) throughout our diffusion training and inference. For spatial normalization layer we use $\gamma \in [0.6, 0.8]$ and for projection layer we use a convolutional layer with kernel size 3 and padding 1. For optimization, we use AdamW (Kingma & Ba, 2014; Loshchilov, 2017) with a constant learning rate of 1×10^{-4} , and a global batch size of 256. During training, we use `bfloat16` mixed precision and `torch.compile` to accelerate training, and gradient clipping and exponential moving average (EMA) to the generative model for stable optimization.

For REPA-E (Leng et al., 2025a) and JiT (Li & He, 2025a) experiments, we use the official open-source implementation. For MeanFlow (Geng et al., 2025) experiments, we received the implementation (including REPA) from original authors and adapt it to introduce the two straightforward changes for iREPA (§4).

Evaluation. For image generation evaluation, we strictly follow the ADM setup (Dhariwal & Nichol, 2021). We report generation quality using Fréchet inception distance (gFID) (Heusel et al., 2017), structural FID (sFID) (Nash et al., 2021), inception score (IS) (Salimans et al., 2016), precision (Prec.) and recall (Rec.) (Kynkäänniemi et al., 2019), measured on 50K generated images. For sampling, we follow REPA (Yu et al., 2024) exactly, using the SDE Euler-Maruyama sampler with 250 steps. For JiT (Li & He, 2025a), we use 50 inference steps following the official implementation.

H MORE DISCUSSION ON RELATED WORK

Classical spatial features in computer vision. Spatial feature extraction has long been fundamental to computer vision. Classical methods like SIFT (Lowe, 1999), HOG (Dalal & Triggs, 2005), SURF (Bay et al., 2006), and ORB (Rublee et al., 2011) providing robust local descriptors for tasks ranging from object detection to image matching. While these handcrafted features excel at capturing local spatial patterns and geometric invariances, recent work in generative modeling has focused almost exclusively on modern self-supervised representations with strong global classification performance, such as DINOv2 (Oquab et al., 2024) and CLIP (Radford et al., 2021). Our findings call for a different perspective: because spatial structure preservation is what governs generation quality, even classical spatial features improve diffusion training when properly aligned. This opens up the full spectrum of spatial feature extractors, from traditional handcrafted descriptors to modern learned representations, as viable alignment targets, provided they maintain strong spatial coherence.

Pretrained visual encoders for generative models. Pretrained visual encoders have supported generative models in several capacities: as discriminators to accelerate GAN convergence (Goodfellow et al., 2020; Sauer et al., 2021; 2022; 2023; Radford et al., 2021), as teachers in adversarial distillation for diffusion models (Sauer et al., 2024; Kang et al., 2024), and more recently as alignment targets. In GANs (Goodfellow et al., 2020), pretrained features have not only improved convergence speed but also enabled scaling to large datasets, as demonstrated by StyleGAN-XL (Sauer et al., 2022) and StyleGAN-T (Sauer et al., 2023) with CLIP (Radford et al., 2021) features. For diffusion models, adversarial distillation leverages pretrained encoders to guide student networks toward higher-fidelity samples, showing clear improvements in perceptual quality. In particular, REPA (Yu et al., 2024) aligns diffusion features with external encoders, demonstrating that representation alignment improves both generation convergence and quality. We build on this direction but shift the emphasis decisively: what matters is the spatial structure these encoders preserve, not their discriminative capability.

Denosing transformers. Transformer architectures have become the dominant backbone for scalable generative modeling, with various formulations including diffusion transformers (Peebles & Xie, 2023) and flow matching variants (Ma et al., 2024). Recent architectures like GenTron (Chen et al., 2023) scale transformers to over 3B parameters for text-to-image synthesis. U-ViT (Bao et al., 2023) demonstrates that pure transformer backbones without U-Net inductive biases can achieve competitive performance. ARDiT (Tang et al., 2024) and DART (Jing et al., 2024) explore autoregressive formulations that combine denoising with sequential generation, enabling flexible trade-offs between quality and speed. Despite these architectural advances, training these models

from scratch remains computationally expensive, often requiring millions of iterations to achieve good generation quality. While representation alignment methods like REPA (Yu et al., 2024) have shown that pretrained features can dramatically accelerate convergence, the mechanism behind this improvement remained unexplained. Our analysis identifies it: the key benefit comes from preserving spatial structure, not from semantic alignment. This explains why certain encoders accelerate training far more than others and gives a principled basis for designing more effective alignment strategies. While (Wang et al., 2025c) propose improvements to REPA training through early stopping, they hypothesize that “REPA predominantly distills global semantic information while leaving structural information untapped.” In contrast, our analysis reveals that spatial structure (not global semantic information) already plays a very significant role in the effectiveness of REPA (§2, §3). Thus, while we indeed find that spatial structure remains underexploited (§4) in REPA, we surprisingly find that majority of improvements in REPA are already coming from introducing an inductive bias for spatial structure (not global information).

Denosing as self-supervised learning task. The connection between denoising and representation learning has been explored from multiple perspectives. Early work by Abstreiter et al. (2021) extended diffusion objectives for representation learning, demonstrating that denoising naturally learns meaningful features. SODA (Hudson et al., 2024) introduces a diffusion model with an information bottleneck to learn compact representations. Chen et al. (2024) deconstruct diffusion models and find that a simple denoising autoencoder suffices for strong self-supervised performance. Wang & He (2025) introduce a dispersive loss to encourage internal representation diversity in diffusion models, improving generative performance without external encoders. Similarly, Jiang et al. (2025) propose Self-Representation Alignment (SRA) to align a diffusion transformer’s latent features across noise levels, providing self-guidance without an auxiliary model. These works establish denoising as a fundamental self-supervised task that naturally encourages learning of robust features. Our findings sharpen this view: when diffusion models are aligned with strong *spatial* representations from self-supervised encoders, both generative and discriminative capabilities improve, pinpointing spatial structure preservation as the key factor in this synergy.

Scaling self-supervised vision encoders. Recent years have seen remarkable progress in scaling self-supervised vision models to unprecedented sizes and datasets. DINOv3 (Siméoni et al., 2025) trains a 7B parameter ViT on 1.7 billion images without labels by aligning representations from different augmentations. WebSSL (Fan et al., 2025) demonstrates that visual models trained on more than 2 billion images can match language-supervised models like CLIP (Radford et al., 2021) on vision-language tasks without language supervision. C-RADIO (Heinrich et al., 2024) combines multiple teacher models through distillation to create versatile encoders that excel across diverse visual domains. I-JEPA (Assran et al., 2023) explores predictive architectures that learn by predicting masked regions in representation space, instead of reconstructing pixels directly. SAM (Ravi et al., 2024) specializes in promptable segmentation through large-scale supervised training. The Perception Encoder family (Bolya et al., 2025; Cho et al., 2025) shows that intermediate features often outperform final representations for dense prediction tasks. While these models are typically evaluated on global tasks like image classification, our work delivers a crucial correction: strong performance on discriminative benchmarks does not translate to better generation quality. What determines the benefit for diffusion training is spatial structure preservation, regardless of classification accuracy. This means the evaluation metrics for self-supervised encoders must be reconsidered for generative applications, where spatial coherence is at least as important as semantic understanding.

I NOTE ON LLM USAGE

We use GPT-5 (OpenAI, 2025) for considering different variations of the spatial structure metrics discussed in the paper §3. Furthermore, all figures in the paper are directly generated from our experiment logs and checkpoints using Claude-Code (Anthropic, 2025). Additionally we use LLM help for searching and formulating relevant work in Appendix H. We use cursor in some parts to help with paper writing.

J ADDITIONAL DISCUSSION ON SPATIAL STRUCTURE METRICS

Spatial structure metrics measure the representation gap. Yu et al. (2024) measure the representation gap with linear probing, using the increase in validation accuracy to explain REPA’s effectiveness. Our spatial structure metrics (SSM) offer a better instrument: they too measure the representation gap and explain improvements across models, but through the quantity that actually matters. As Figure 17 shows, representation alignment closes the gap between the SSM performance of a pretrained encoder like DINOv2 and that of the diffusion features.

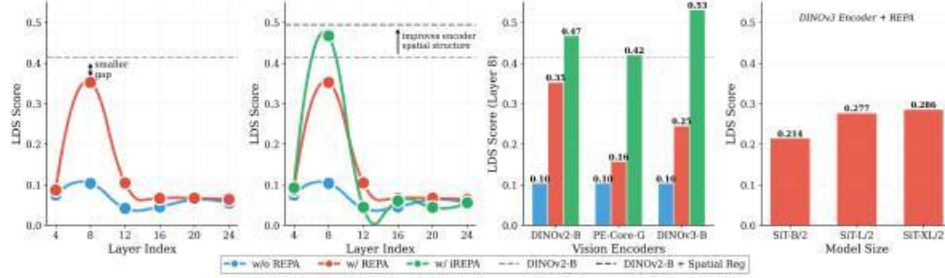


Figure 17: Representation alignment as bridging the spatial feature gap. a) Representation alignment can be understood as bridging the spatial feature gap between diffusion and vision encoder patch features. b) iREPA (§4) accentuates the vision encoder’s spatial features (at the cost of some global information), directly improving generation performance. c) iREPA improves performance across every vision encoder. d) The spatial structure improvements scale with model size. All results are reported with SiT-XL/2 at 400k iterations.

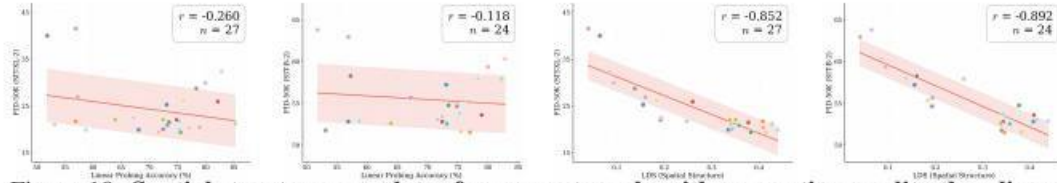


Figure 18: Spatial structure correlates far more strongly with generation quality than linear probing. Correlation analysis across 27 vision encoders. **Left two:** Linear probing accuracy vs FID is only weakly predictive, for SiT-XL-2 (Pearson $r = -0.26$) and SiT-B-2 ($r = -0.12$). **Right two:** Spatial structure (LDS) vs FID is strongly predictive, for SiT-XL-2 ($r = -0.85$) and SiT-B-2 ($r = -0.89$). See Figure 10 in Appendix for detailed plots with encoder labels.

K ADDITIONAL ANECDOTAL COMPARISONS

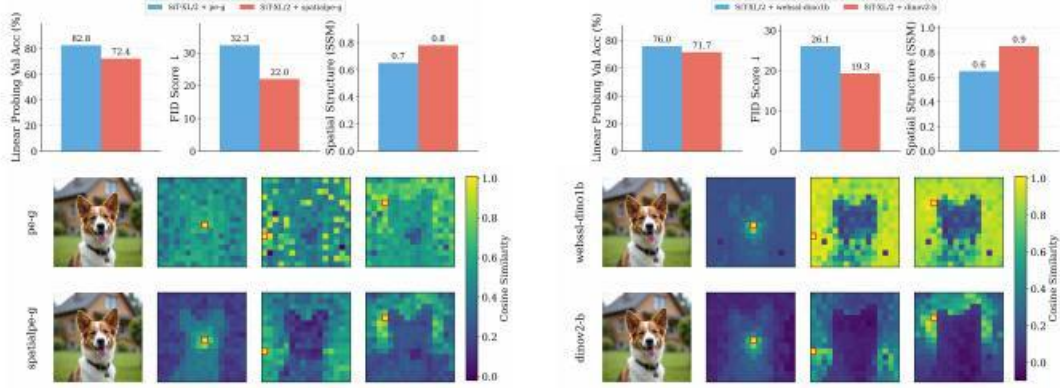


Figure 19: **Motivating examples from recent SSL encoders — spatial structure matters.** **Top row:** The comparison exposes a clear inverse relationship between ImageNet accuracy and generation quality. **Left:** PE-g has higher accuracy (82.8% vs 72.4%) but worse FID (32.3 vs 22.0) than Spatial-PE-g, and much lower spatial structure (LDS: 0.1 vs 0.4). **Right:** WebSSL-dino1b has higher accuracy (76.0% vs 71.7%) but worse FID (26.1 vs 19.3) than DINOv2-b, and weaker spatial structure (LDS: 0.2 vs 0.4). **Bottom row:** The spatial cosine similarity visualizations make the cause explicit: the encoders that generate better (Spatial-PE-g, DINOv2-b) retain clear spatial coherence, while those optimized for classification (PE-g, WebSSL) lose spatial structure.

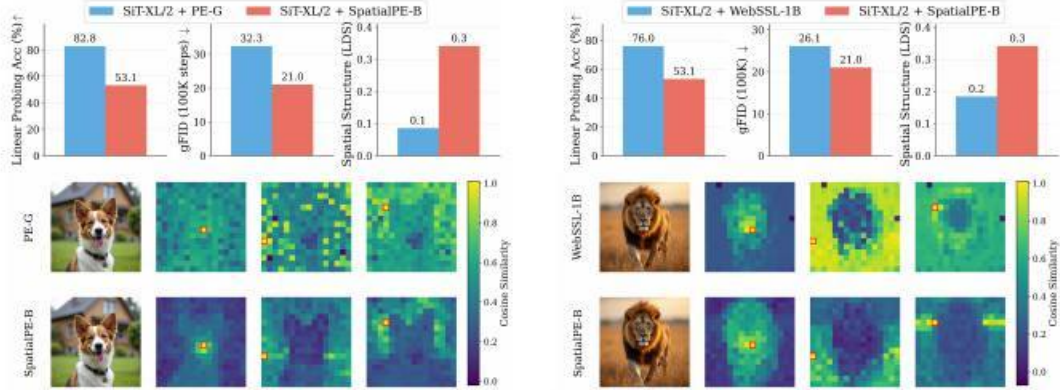


Figure 20: **Motivating examples — spatial structure matters.** **Top:** The comparison exposes a clear inverse relationship between ImageNet accuracy and generation quality. **Left:** PE_{core}-G, despite far higher validation accuracy (82.8% vs. 53.1%), generates *worse* than PE_{Spatial}-B (Bolya et al., 2025). **Right:** WebSSL-1B (Fan et al., 2025) tells the same story: much better global performance (76.0% vs. 53.1%), but worse generation. **Bottom:** Spatial structure, not global performance, is the reliable predictor of generation quality. See §3 for the spatial structure metric. All results reported at 100K using SiT-XL/2 and REPA.

L FURTHER DISCUSSION

Role of Spatial Normalization We visualize the impact of spatial normalization on vision encoder representations. Vision encoders carry significant global components, or overlays, that suppress the contrast between spatial tokens: tokens in one semantic region become highly correlated with tokens in unrelated regions (e.g., background), erasing the spatial distinctiveness of features. Spatial normalization removes this global overlay and restores contrast between spatial tokens, letting the model preserve local spatial structure while stripping away the global information that interferes with generation quality.

The figures below make this effect concrete across several examples, showing how spatial normalization reshapes the feature representations to emphasize spatial structure:

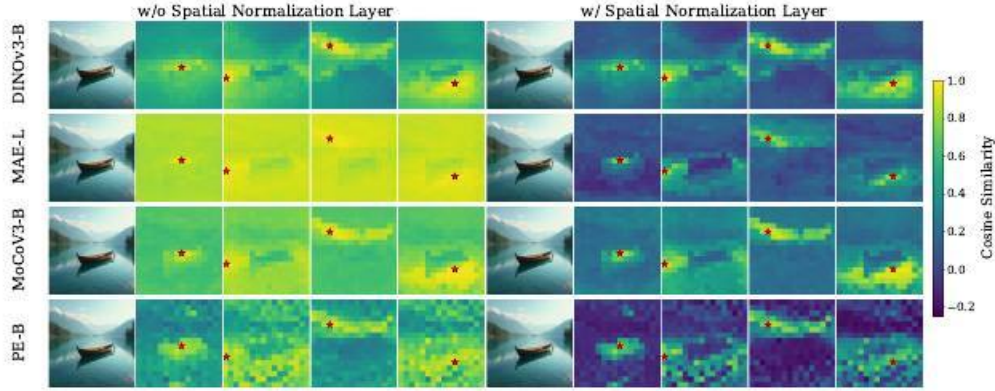


Figure 21: Visualizing the impact of spatial normalization (Example 1). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components drive high correlations across unrelated regions. With spatial normalization (right), local spatial structure is sharpened and global interference removed, producing distinctly cleaner semantic boundaries.

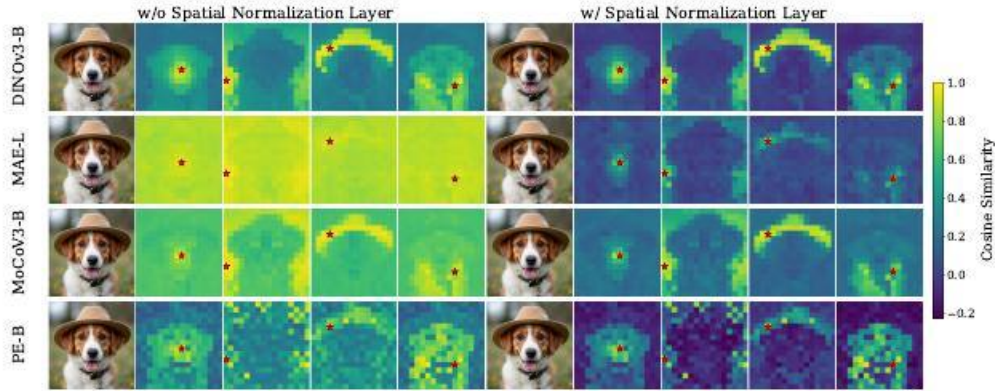


Figure 22: Visualizing the impact of spatial normalization (Example 3). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components drive high correlations across unrelated regions. With spatial normalization (right), local spatial structure is sharpened and global interference removed, producing distinctly cleaner semantic boundaries.

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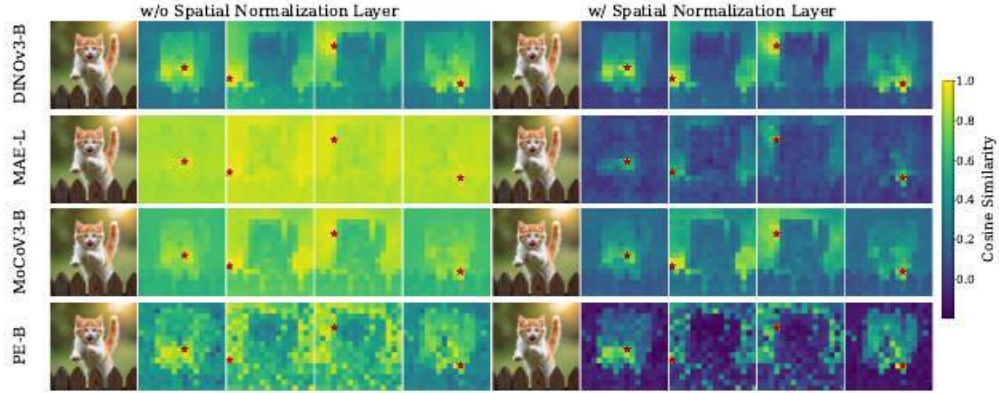


Figure 23: Visualizing the impact of spatial normalization (Example 2). The feature similarity maps show how spatial normalization sharpens spatial contrast. The original features (left) carry a global overlay that blurs the distinction between foreground and background. After normalization (right), spatial tokens become locally coherent, with clear separation between semantic regions.

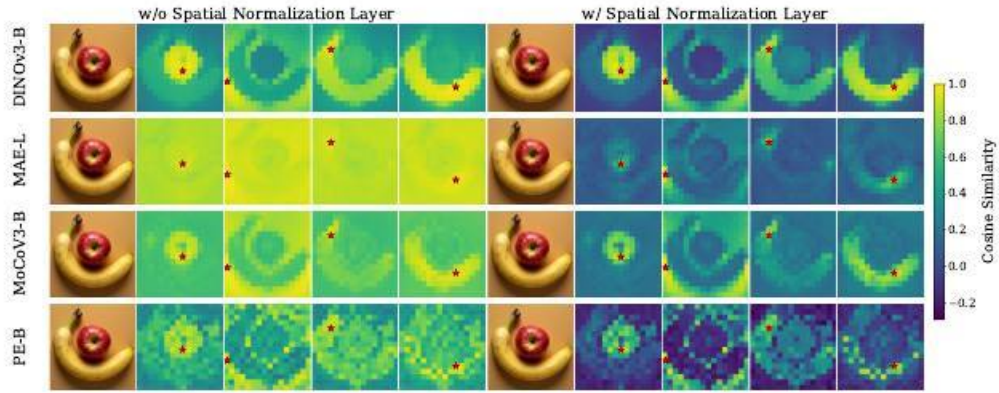


Figure 24: **Visualizing the impact of spatial normalization** (Example 4). The heatmaps show token similarity patterns before and after spatial normalization. Without normalization (left), global components drive high correlations across unrelated regions. With spatial normalization (right), local spatial structure is sharpened and global interference removed, producing distinctly cleaner semantic boundaries.

M ADDITIONAL RESULTS

M.1 ADDITIONAL RESULTS AT HIGHER RESOLUTIONS

Setup. We further test the proposed spatial improvements at higher resolution on ImageNet-512 (Deng et al., 2009). We follow the same setup as REPA (Yu et al., 2024), reporting results across different pretrained encoders (DINOv2, DINOv3, WebSSL, PE, etc.). All results use SiT-XL and SiT-B with 50 NFE at inference w/o classifier free guidance. Fig. 25 compares REPA before and after applying the spatial improvements (iREPA). The spatial improvements generalize cleanly to higher resolution, with consistent gains across every pretrained encoder tested (DINOv2, DINOv3, WebSSL, PE, etc.).

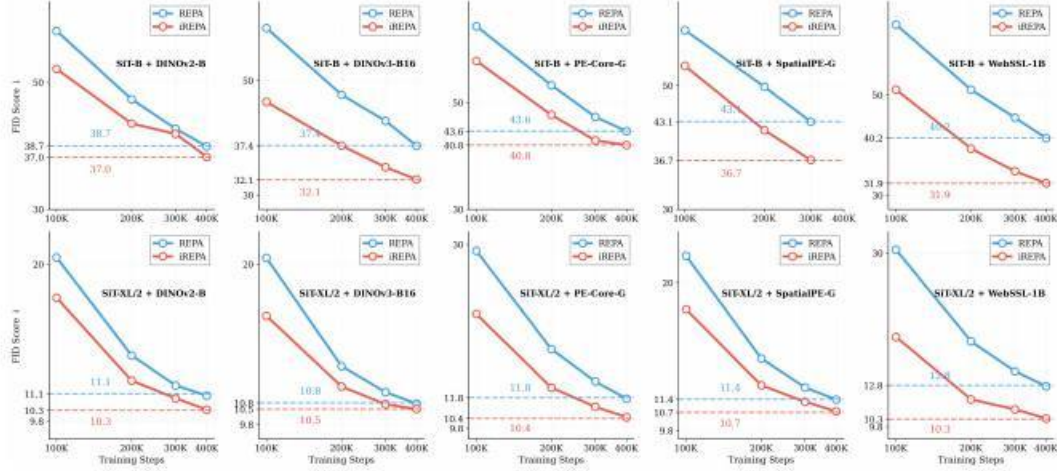


Figure 25: **Convergence results at 512x512 resolution.** Accentuating spatial features consistently improves convergence speed at both resolutions, Imagenet 256 (Figure 7) and Imagenet 512 (above).

M.2 ADDITIONAL RESULTS ON MULTIMODAL T2I TASKS

Setup. To establish that the proposed spatial improvements generalize beyond ImageNet, we run extensive experiments on multimodal T2I tasks. Following REPA (Yu et al., 2024), we adopt MMDiT (Esser et al., 2024) as the diffusion backbone and apply REPA with various pretrained vision encoders (DINOv2, CLIP, WebSSL, PE, etc.). As in REPA (Yu et al., 2024), all models are trained for 150K steps with a batch size of 256 and evaluated using an ODE sampler with 50 NFE. Fig. 26 compares REPA before and after applying the spatial improvements (iREPA). The spatial improvements carry over to multimodal T2I tasks as well, with consistent gains across every pretrained encoder tested (DINOv2, CLIP, WebSSL, PE, etc.).

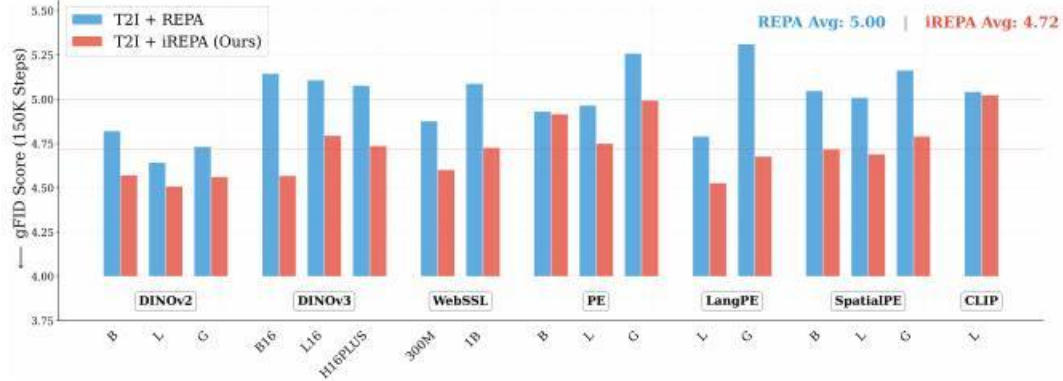


Figure 26: **Text-to-image generation across encoder variants.** Accentuating the transfer of spatial features from the target representation to the diffusion features consistently improves convergence speed for both imagenet (refer §4) and multimodal T2I tasks (above), with consistent gains across every pretrained encoder tested (DINOv2, CLIP, WebSSL, PE, etc.).

M.3 CORRELATION ANALYSIS WITHOUT OUTLIERS (MoCOv3-L AND MAE-L)

We repeat the correlation analysis from Section 3 after removing MoCOv3-L and MAE-L, which were identified as outliers. Figure 28 shows the updated correlations across different model sizes. Our conclusion is unchanged: spatial structure remains far more predictive of generation performance than linear probing accuracy. Tellingly, once the outliers are removed, linear probing turns *positively* correlated with gFID, meaning higher linear probing accuracy is associated with *worse* generation. This directly reinforces the observations in Sec. 2, where target representations with higher global semantic performance (linear probing accuracy) tend to show similar or worse generation performance under representation alignment (REPA).

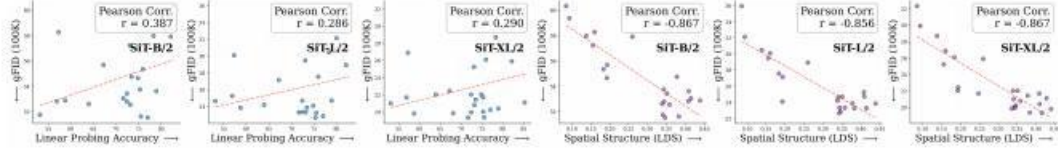


Figure 27: **Correlation analysis without outliers.** Across all model sizes (B, L, XL), and even after removing the outliers (MoCOv3-L and MAE-L), spatial structure remains far more predictive (Pearson $|r| > 0.85$) of generation performance than linear probing accuracy (Pearson $|r| < 0.38$). Tellingly, once the outliers are removed, linear probing turns positively correlated with gFID, meaning higher linear probing accuracy is associated with worse generation. This directly reinforces the observations in Sec. 2, where target representations with higher global semantic performance (linear probing accuracy) tend to show similar or worse generation performance under representation alignment (REPA).

M.4 ADDITIONAL QUALITATIVE RESULTS



Figure 28: **Additional Qualitative Results** comparing generation outputs before and after applying the spatial improvements with REPA. All results are reported using PE-G (Bolya et al., 2025) as the pretrained vision encoder, 400K steps (80 epochs) and with classifier-free guidance scale of 4.0. The visual results mirror the quantitative gains (§4): the spatial improvements (iREPA) clearly improve the visual quality and coherence of the generated outputs.

M.5 ADDITIONAL RESULTS WITH ALTERNATIVE EVALUATION METRICS

In addition to traditional evaluation metrics (Inception Score, FID, sFID, Precision, Recall), we also verify the robustness of the proposed findings with alternative evaluation metrics for generation quality (Stein et al., 2023; Yang et al., 2023; Kynkäänniemi et al., 2023; Jayasumana et al., 2024). In particular, we use the CMMD metric (Jayasumana et al., 2024) with the PyTorch implementation from (Paul, 2024). Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. We next verify the robustness of the key findings from both §3 and §4 with the CMMD metric.

Spatial structure correlates much higher with generation performance than linear probing. To study robustness of analysis from §3, we repeat the large-scale correlation analysis across different vision encoders with CMMD metric (Jayasumana et al., 2024). Results are shown in Figure 29. All results are reported using SiT-B/2 (100K steps) and REPA. All spatial metrics show much higher correlation (Pearson $|r| > 0.88$) with generation performance (CMMD) than linear probing (Pearson $|r| = 0.074$) — demonstrating that key empirical findings from §3 are robust to the choice of evaluation metric (FID or CMMD).

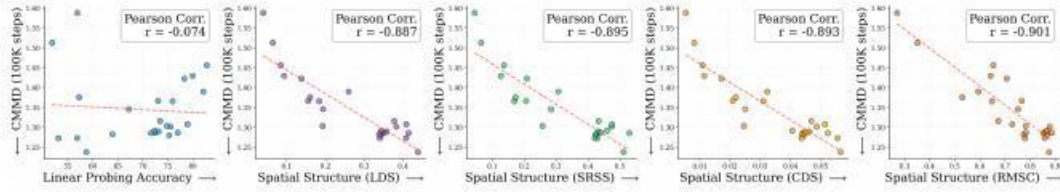


Figure 29: Correlation analysis with CMMD metric. We repeat the correlation analysis from §3 with CMMD metric (Jayasumana et al., 2024) instead of gFID. All results are reported using SiT-B/2 (100K steps) and REPA. Every spatial metric correlates strongly (Pearson $|r| > 0.88$) with generation performance (CMMD), while linear probing barely correlates (Pearson $|r| = 0.074$). The key findings from §3 thus hold regardless of the evaluation metric.

Accentuating the transfer of spatial features improves generation quality. We likewise re-examine the analysis from §4 under the CMMD (Jayasumana et al., 2024) metric alongside the traditional evaluation metrics (gFID, sFID, IS, etc.). Results are shown in Table 8. Across every vision encoder, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and the traditional evaluation metrics (IS, FID, sFID, Prec., Recall), with consistent improvements both with and without classifier-free guidance (CFG).

Vision Encoder	Steps	w/o CFG						w/ CFG					
		CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑	CMMD↓	IS↑	FID↓	sFID↓	Prec.↑	Rec.↑
DINOv2-B	100K	0.702	69.20	19.3	5.89	0.64	0.61	0.529	157.2	6.35	5.91	0.77	0.54
+iREPA	100K	0.652	77.92	16.9	6.26	0.66	0.61	0.484	179.3	5.15	6.23	0.78	0.54
DINOv2-B	400K	0.455	127.4	7.76	5.06	0.70	0.66	0.320	263.0	1.98	4.60	0.80	0.61
+iREPA	400K	0.438	128.6	7.52	4.89	0.71	0.65	0.310	268.8	1.93	4.59	0.80	0.60
DINOv3-B	100K	0.749	63.64	21.4	6.14	0.63	0.60	0.571	144.0	7.57	6.09	0.76	0.53
+iREPA	100K	0.651	78.79	16.2	6.14	0.66	0.61	0.481	181.9	4.87	6.10	0.78	0.55
DINOv3-B	400K	0.474	126.7	8.10	5.06	0.70	0.66	0.336	261.2	1.99	4.58	0.80	0.61
+iREPA	400K	0.441	132.9	7.13	4.93	0.71	0.66	0.314	272.4	1.89	4.58	0.80	0.60
WebSSL-1B	100K	0.825	53.87	25.5	6.57	0.61	0.59	0.622	124.0	9.59	6.37	0.76	0.52
+iREPA	100K	0.653	77.47	16.6	6.18	0.66	0.61	0.484	177.6	5.09	6.11	0.79	0.54
WebSSL-1B	400K	0.512	116.9	9.39	5.14	0.70	0.64	0.354	250.5	2.24	4.61	0.81	0.58
+iREPA	400K	0.445	130.8	7.48	4.91	0.70	0.65	0.311	271.6	1.90	4.58	0.80	0.61
PE-Core-G	100K	0.922	42.74	32.3	6.70	0.57	0.59	0.714	97.2	14.1	6.56	0.71	0.53
+iREPA	100K	0.697	75.01	18.1	6.03	0.64	0.61	0.525	176.8	5.66	6.08	0.77	0.54
PE-Core-G	400K	0.525	109.4	10.4	5.00	0.69	0.64	0.366	238.4	2.44	4.57	0.81	0.59
+iREPA	400K	0.458	132.0	7.78	5.02	0.70	0.65	0.322	275.4	1.93	4.59	0.80	0.61
CLIP-L	100K	0.790	54.07	25.2	6.65	0.61	0.60	0.605	124.5	9.77	6.59	0.75	0.52
+iREPA	100K	0.657	74.46	17.8	6.33	0.65	0.61	0.489	172.0	5.67	6.33	0.78	0.54
CLIP-L	400K	0.487	117.8	8.97	4.98	0.70	0.65	0.339	258.7	2.15	4.64	0.81	0.59
+iREPA	400K	0.442	130.1	7.42	5.03	0.71	0.65	0.307	271.8	1.96	4.64	0.80	0.61

Table 8: **Additional results with alternative evaluation metrics.** We provide additional results with the CMMD metric (Jayasumana et al., 2024). All results are reported using SiT-XL/2, 256 batch size and traditional REPA as baseline. We adopt the pytorch implementation from (Paul, 2024) for computing the CMMD metric. Standard reference set from OpenAI ADM evaluation suite (Dhariwal & Nichol, 2021) is used as reference images. Across every vision encoder, accentuating the transfer of spatial features (iREPA) improves convergence speed under both CMMD and traditional evaluation metrics (IS, FID, etc.). The key empirical findings from §4 therefore hold regardless of the evaluation metric.

M.6 SPATIAL IMPROVEMENTS WITH SAM2

Table 9 and Figure 30 study the impact of spatial normalization (§4) on SAM2-S (46M) encoder features². For SAM2, spatial normalization helps but only marginally, and our framework predicts exactly why: spatial normalization works by removing the global component (mean of patch tokens) to sharpen spatial contrast, yet SAM2 already carries little to no global information (validation accuracy < 24%), leaving little for normalization to remove (see Figure 30). The small gain is thus a confirmation, not a counterexample, of our account.

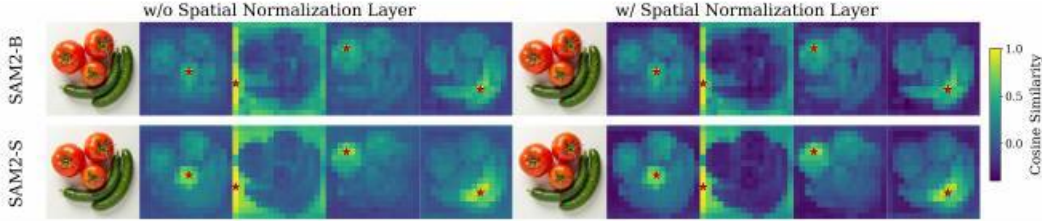


Figure 30: **Impact of spatial normalization on SAM2.** For SAM2, the spatial normalization layer helps but only marginally, and our framework predicts exactly why. Spatial normalization (§4) sharpens spatial contrast by removing the global component (mean of patch tokens); since SAM2 already carries little to no global information (validation accuracy < 24%), there is little for it to remove, so the contrast improves only slightly.

Vision Encoder	Steps	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
SAM2-S	100K	50.69	25.32	6.52	0.631	0.588
+iREPA	100K	53.14	24.52	6.28	0.629	0.591
SAM2-S	400K	110.77	9.54	4.93	0.699	0.638
+iREPA	400K	114.62	9.10	4.89	0.704	0.640

Table 9: **Impact of spatial improvements on SAM2.** All results are reported with SiT-XL/2, without classifier-free guidance, SAM2-S (46M) as vision encoder and with traditional REPA as the baseline.

²Note that similar to (Bolya et al., 2025), we use the intermediate output of vision encoder for SAM2 features and not the mask logits. As shown in (Bolya et al., 2025), while mask logits lead to sharper spatial maps, the mask logits itself are not suitable as a target representation.

M.7 ADDITIONAL RESULTS WITH FULL-FINETUNING ACCURACY INSTEAD OF LINEAR PROBING ACCURACY

Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. Our conclusion does not hinge on the choice of global-information proxy. We repeat the correlation analysis from §3 using validation accuracy after full-finetuning³ in place of linear probing. Results are shown in Fig. 31. All results are reported using SiT-XL/2 (100K steps) and REPA. For full-finetuning validation accuracy, instead of linear probing, we perform full-finetuning of vision encoder with learning rate of $5e-5$, warmup ratio of 0.1, and total of 30 epochs. Every spatial metric is far more predictive of generation performance than full-finetuning accuracy. In fact, gFID shows a weak *positive* correlation with full-finetuning validation accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID, i.e., worse generation. This again reinforces the observations in §2, where target representations with higher global semantic performance tend to show similar or worse generation performance under representation alignment (REPA).

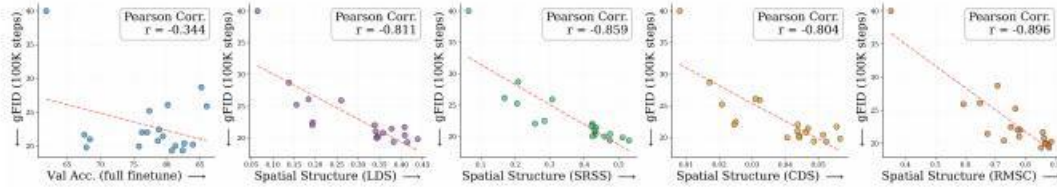


Figure 31: Correlation analysis with full-finetuning accuracy instead of linear probing accuracy. We repeat the correlation analysis from §3 with the validation accuracy after full-finetuning instead of linear probing. All results are reported using SiT-XL/2 (100K steps) and REPA. Every spatial metric is far more predictive of generation performance than full-finetuning accuracy. In fact, gFID shows a weak positive correlation with full-finetuning validation accuracy (Pearson $r = 0.317$): higher validation accuracy is associated with higher gFID, i.e., worse generation. This again reinforces the observations in §2, where target representations with higher global semantic performance tend to show similar or worse generation performance under representation alignment (REPA).

³Please note that while the final findings remain similar, in context of REPA linear probing might be more accurate for estimating global information in “pretrained” vision encoders. This is because REPA uses the “pretrained” encoder features themselves for regularization. Finetuning the encoder itself can impact the amount of global information. Same as REPA (Yu et al., 2024), we therefore use linear probing accuracy as default for measuring global information.

M.8 ADDITIONAL RESULTS ON PIXEL-SPACE DIFFUSION MODELS

Method	#Params	IS $\uparrow$	FID $\downarrow$	sFID $\downarrow$	Prec. $\uparrow$	Rec. $\uparrow$
PixelFlow (Chen et al., 2025)	459M	24.67	54.33	9.71	-	0.58
PixDDT (Wang et al., 2025b)	434M	36.24	46.37	17.14	-	0.63
PixNerd (Wang et al., 2025a)	458M	43.01	37.49	10.65	-	0.62
DeCo w/o $\mathcal{L}_{\text{FreqFM}}$	426M	46.44	34.12	10.41	-	0.64
DeCo + REPA (Ma et al., 2025b)	426M	48.35	31.35	9.34	-	0.65
JiT (Li & He, 2025a)	459M	29.37	49.06	11.21	0.40	0.62
JiT+iREPA (Ours)	459M	50.72	29.19	9.42	0.51	0.65

Table 10: **Generation performance of pixel-space diffusion models.** With its two simple modifications, iREPA combined with JiT (Li & He, 2025a) surpasses recent state-of-the-art pixel-space diffusion methods such as DeCo (Ma et al., 2025b). All results are reported with 200K training iterations with a batch size of 256 and evaluated using 50-step Euler sampling without classifier-free guidance. Results for DeCo are obtained directly from (Ma et al., 2025b).